nature microbiology

LETTERS

https://doi.org/10.1038/s41564-020-0686-0

Check for updates

Genetically engineered control of phenotypic structure in microbial colonies

Philip Bittihn[1,2,5], Andriy Didovyk[1,6], Lev S. Tsimring[1,2] ✉ and Jeff Hasty[1,2,3,4] ✉

Rapid advances in cellular engineering[1,2] have positioned synthetic biology to address therapeutic[3,4] and industrial[5] problems, but a substantial obstacle is the myriad of unanticipated cellular responses in heterogeneous real-world environments such as the gut[6,7], solid tumours[8,9], bioreactors[10] or soil[11]. Complex interactions between the environment and cells often arise through non-uniform nutrient availability, which generates bidirectional coupling as cells both adjust to and modify their local environment through phenotypic differentiation[12,13]. Although synthetic spatial gene expression patterns[14–17] have been explored under homogeneous conditions, the mutual interaction of gene circuits, growth phenotype and the environment remains a challenge. Here, we design gene circuits that sense and control phenotypic structure in microcolonies containing both growing and dormant bacteria. We implement structure modulation by coupling different downstream modules to a tunable sensor that leverages *Escherichia coli*'s stress response and is activated on growth arrest. One is an actuator module that slows growth and thereby alters nutrient gradients. Environmental feedback in this circuit generates robust cycling between growth and dormancy in the interior of the colony, as predicted by a spatiotemporal computational model. We also use the sensor to drive an inducible gating module for selective gene expression in non-dividing cells, which allows us to radically alter population structure by eliminating the dormant phenotype with a 'stress-gated lysis circuit'. Our results establish a strategy to leverage and control microbial colony structure for synthetic biology applications in complex environments.

In natural habitats, bacteria often live in colonies with prominent internal structures and under strongly non-uniform environmental conditions[18,19]. Environmental and biological heterogeneity are inseparably linked by complex feedback processes, because bacterial metabolism and growth dynamics are modulated by the local environment that they themselves create by consuming and secreting chemicals[13]. As a result, gene expression patterns depend on the location of the individual cell within the colony, and the specific physical and chemical local environment[20], mediated, for example, by the stress response[21]. Therefore, the behaviour of synthetic microbes designed under laboratory conditions can be dramatically impacted when exposed to real-world environments[22].

One of the most prominent drivers of spatial heterogeneity is inhomogeneous nutrient availability, which naturally develops in sufficiently large, diffusion-limited environments colonized by bacterial populations. Although the links between nutrients, growth rate and gene expression profiles have been studied in well-mixed environments[23], and gene expression from synthetic modules has been observed in non-standard phenotypes[24], this information has not been leveraged to create gene circuits specifically designed to operate in structured bacterial populations with multiple coexisting phenotypes. In this study, we use microfluidic technology to create well-defined model environments in which bacterial populations reproducibly develop phenotypic heterogeneity due to diffusion-limited nutrient availability. We design basic modules that (1) respond to the phenotypic changes, (2) modulate cellular growth to change nutrient consumption and (3) limit expression from an inducible promoter to a specific phenotype. We then use them to construct two proof-of-principle gene circuits: one that can alter nutrient gradients in a feedback-like manner, and the other to selectively eliminate non-growing cells from the population.

In our idealized confined environment, deep side-traps protrude from nutrient supply channels that continuously deliver fresh medium to one side of the growing colony while washing away spillover cells (Fig. 1a, and see Supplementary Figs. 1 and 2). When loaded into the device, *Escherichia coli* colonies grew to fill the traps. Cells deprived of nutrients at the rear of the traps switched to a non-dividing phenotype resembling the small, round morphology in stationary-phase batch cultures[25] (Fig. 1b, and see Extended Data Fig. 1 and Supplementary Video 1), indicating that nutrient supply is inherently diffusion limited[26]. In the rich medium used in these experiments (see Methods), the steady-state colony structure featured a sharp boundary (at ~2 h) between rod-shaped dividing and small non-dividing cells (Fig. 1c). This emergent structure served as the reference state for our further investigation.

For growth-state coupled regulatory circuits, downstream modules need to be able to detect the current state of the cell. To this end, we constructed a 'stress sensor module' that is activated on transition into stationary phase, when the alternative/stationary phase sigma factor (σ^S) becomes active. Our sensor is based on the σ^S-dependent promoter pOsmY (see Methods) which responds to metabolic stress (nutrient deprivation) and the response of which can be modulated by NaCl concentration[27]. When it was used to drive green florescent protein (GFP) expression (Fig. 1d), we observed a sharp increase in fluorescence on entry into stationary phase in batch culture, which could be further amplified by increased osmolarity, providing an experimental knob for tuning expression levels of pOsmY (Fig. 1e). Sensor activation in microfluidics occurred on initial microcolony growth arrest, and in dividing cells close to the no-growth boundary when exposed to higher osmolarity, consistent with batch culture results (Fig. 1f, and see Extended Data Fig. 3a and Supplementary Video 2).

[1]BioCircuits Institute, University of California, San Diego, La Jolla, CA, USA. [2]The San Diego Center for Systems Biology, La Jolla, CA, USA. [3]Department of Bioengineering, University of California, San Diego, La Jolla, CA, USA. [4]Molecular Biology Section, Division of Biological Sciences, University of California, San Diego, La Jolla, CA, USA. [5]Present address: Max Planck Institute for Dynamics and Self-Organization, Göttingen, Germany. [6]Present address: Vertex Pharmaceuticals, San Diego, CA, USA. ✉e-mail: ltsimring@ucsd.edu; hasty@bioeng.ucsd.edu

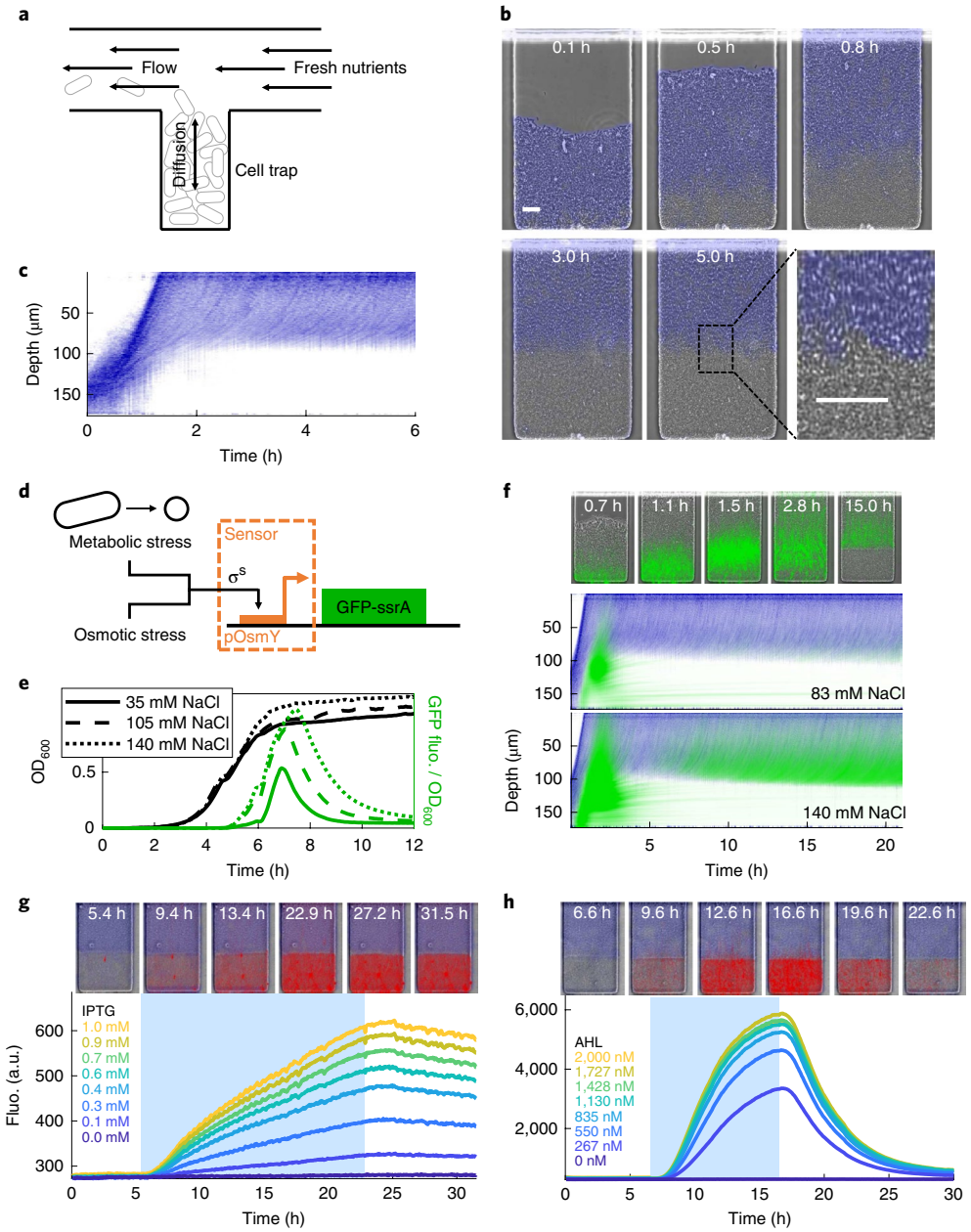


Fig. 1 | Microfluidic platform for observing *E. coli* growth patterns in spatially heterogeneous microcolonies. a, Schematic illustrating a trap in our microfluidic device. The flow rate in the nutrient supply channel is ≈100 μm s^{-1}. **b**, Time-lapse microscopic images of a single microfluidic trap showing development of phenotypic heterogeneity shortly after cell loading (see Supplementary Video 1). Blue colour indicates regions of actively growing *E. coli* (based on IIDs; see Methods). Scale bars (top left and bottom right images), 15 μm. **c**, Kymograph of IIDs (blue) averaged along the horizontal dimension of the trap. **d**, The stress sensor consists of the σ^S-responsive pOsmY promoter driving ssrA-tagged GFP to visualize sensor activation (**e**,**f**). **e**, Time courses of optical density (OD, black) and fluorescence (fluo., green) in plate reader experiments display transient sensor activation that depends on the level of basal osmotic stress (NaCl). **f**, Time-lapse images (for 140 mM NaCl; see Supplementary Video 2) and kymographs of microfluidic experiments showing NaCl-dependent transient stress sensor activation (green) on initial growth arrest and later persistent activation in dividing cells close to the no-growth boundary. Blue colour indicates IIDs (see Methods). **g**, Protein expression in growth-arrested *E. coli* measured by induction of IPTG-inducible pLlacO-1 driving RFP. In the time-lapse images, red indicates RFP fluorescence and blue marks the regions of growth. Plotted time traces represent average RFP fluorescence in the growth-arrested region of the trap for different IPTG concentrations present during the induction window (blue shaded area). Expression in growth-arrested cells was also observed with pLuxI and AHL (see Extended Data Fig. 3c). a.u., arbitrary units. **h**, The same for pLuxI-driven RFP which is ssrA tagged and targeted for enzymatic degradation by ClpXP.

As another prerequisite for phenotype-specific expression, we sought to determine whether metabolism was still active enough in growth-arrested bacteria to support the functioning of synthetic constructs. To monitor protein expression, we transformed the cells with a construct expressing red fluorescent protein (RFP) from a LacI-repressed pLlacO-1 promoter[28]. Isopropyl β-*d*-1-thiogalactopyranoside (IPTG)-dependent RFP expression was observed for more than 15 h in dormant cells (Fig. 1g). Similar results were obtained for the pLuxI promoter on induction with *N*-(3-oxohexanoyl)-homoserine lactone (AHL; see Extended Data

Fig. 3c), consistent with long-term gene expression previously seen in stationary cells[24]. For both promoters, the decline of fluorescence on cessation of induction was minimal due to the absence of dilution by cell growth. We tested whether enzymatic degradation by the unfoldase-protease ClpXP[29] was also long lived in dormant cells. Targeting RFP for degradation by adding an ssrA-degradation tag led to a rapid decline of fluorescence on removal of the inducer (Fig. 1h), indicating that non-dividing cells support both long-term expression and degradation in situ.

Next, we looked for a way to actively modulate the phenotypic structure within the microcolonies by artificially modifying their growth state. Comparing traps with depths of 170 µm and 140 µm—the latter containing about half as many non-dividing cells as the former—we observed very similar growth patterns relative to the mouth of the trap (see Extended Data Fig. 2a,b), indicating that non-dividing cells do not actively affect the phenotypic structure. Therefore, we focused our attention on modulating cell growth of the dividing cells.

As an actuator module, we constructed a fluorescent growth repression protein (GRP) by fusing a portion of the OsmY protein with GFP (see Methods) and ssrA tagged it to eliminate toxicity at basal expression levels. When expressed from an IPTG-inducible promoter, GRP-ssrA substantially reduced and ultimately stopped cell growth (Fig. 2a). In the microcolonies growing within the microfluidic traps, growth rate reduction of dividing cells allowed nutrients to diffuse deeper into the traps. Pulsed expression of GRP-ssrA reversibly modulated the phenotypic structure by slowing growth in the front of the trap, causing previously dormant cells in the back to temporarily resume growth (Fig. 2b and see Supplementary Video 3).

We combined the actuator module with our tunable stress sensor module (see Fig. 1d) to reduce growth in the cells experiencing metabolic stress, thereby facilitating nutrient diffusion into the trap (Fig. 2c). This circuit resembles a negative feedback loop, where a gene product suppresses its own production, with the distinction that the feedback loop in our system is mediated by environmental diffusion. A computational reaction–diffusion–advection model based on mass-action kinetic equations for our sensor and actuator modules (see Extended Data Fig. 5), coupled with the equation for the nutrients (see Supplementary Discussion 2), predicted that oscillations should be observed when the stress sensor module is tuned to sufficiently high expression levels (Fig. 2d,e and see Extended Data Fig. 5d). Experimentally, these oscillations were observed in a layer of cells close to the no-growth boundary as regular repetitive modulations of both GRP fluorescence and growth (see Supplementary Video 4; Fig. 2f,g) for sufficiently high NaCl concentrations (above approximately 40 mM). The dormant layer of cells behind the oscillatory region remained undisturbed and supported RFP expression (Fig. 2h). Consistent with the theoretical prediction (Fig. 2e and see Extended Data Fig. 5d), no oscillations were observed in the absence of NaCl, when pOsmY expression levels were too low to achieve a substantial effect on growth (see Extended Data Fig. 6a). However, once induced, the period was insensitive to the sensor induction level determined by NaCl (see Extended Data Fig. 6b).

Although our actuator module has the ability to reactivate some dormant cells in the back of the trap (Fig. 2b), it can only do so by artificially slowing growth in other parts of the population, redistributing a limited amount of nutrients. However, this is not a viable strategy if phenotypic heterogeneity and altered metabolism themselves are undesirable. Instead, it would be necessary to limit the population to a size that can be sustained by the environmental conditions. Pursuing this alternative strategy, we devised an inducible gene circuit that would eliminate stationary-phase cells and restore population homogeneity in this way. We used phage φX174 lysis protein E, which had been used previously for population control[8,30,31]. However, expressing the lysis protein from the standard pLuxI promoter could not selectively eliminate dormant cells without severely affecting dividing cells (see Supplementary Discussion 1, Extended Data Fig. 7 and Supplementary Video 5).

We therefore needed a module that would limit the expression of a target gene specifically to growth-arrested cells and thus avoid lysis protein synthesis in dividing cells. To design this 'gating module', we removed the LuxR transcription factor, which activates the pLuxI promoter in the presence of AHL, from the standard pLuxI cassette, and instead expressed it from the pOsmY stress-sensor promoter (Fig. 3a), enabling pLuxI induction by AHL only once cells enter the growth-arrested state. In the presence of the gating module, LuxR and target gene levels should be reduced in dividing cells and increased in growth-arrested cells compared with constitutive LuxR expression in the standard ungated pLuxI promoter cassette (Fig. 3b).

To test this design, we placed RFP as the target gene under the control of the gating module and tagged it with a weak ssrA-degradation tag ('ssrA-AAV'; see Methods) to prevent accumulation from leaky expression in dormant cells. We defined 'gating fidelity' as the ratio of the maximum fluorescence after the onset of stationary phase to the fluorescence at a specific low optical density (OD) in batch culture (see Methods and Extended Data Fig. 8a). We observed a greater than tenfold increase in gating fidelity (Fig. 3c) compared with non-gated expression, together with a stronger response at much lower AHL levels both in batch culture (see Extended Data Fig. 8b) and in microfluidics (Fig. 3d). Thus, the gating module was effective in increasing expression in growth-arrested cells relative to dividing cells. This growth-state specificity was maintained across different growth media (see Extended Data Fig. 8d,e). A theoretical model for the gating circuit, constructed using the characteristics of the pOsmY promoter from our sensor-actuator circuit (see Supplementary Discussion 2), confirmed that this dramatic improvement in gating fidelity is consistent with the growth-state-specific expression of LuxR (see Extended Data Fig. 9a,b,c). By reducing LuxR expression during exponential growth, the gating module was also able to improve exponential-phase growth rate compared with constitutive LuxR expression (see Supplementary Fig. 3). In addition, induction time—the time to reach 20% of the final fluorescence—in non-dividing cells was substantially reduced compared with the standard (non-gated) divergent-expression pLuxI cassette. The induction time was constant across all AHL concentrations (see Extended Data Fig. 8c), in contrast to the strong dependence on the AHL concentration in the standard pLuxI cassette, where it was possibly caused by the positive autoregulation of LuxR[32].

Placing lysis gene *E* tagged with the same ssrA-AAV-degradation tag as the target gene under the control of the gating module we obtained a stress-gated lysis circuit (SGLC). We assessed the circuit's behaviour in batch culture by analysing growth at different induction levels of AHL and comparing it with ungated expression of the lysis gene from the standard pLuxI expression cassette. Replacing RFP by the lysis gene in our parameterized theoretical model for the gating circuit, we predicted that, in contrast to the ungated lysis circuit, the preferential expression of LuxR during stationary phase in the SGLC would minimize leaky lysis protein expression during exponential phase, while allowing for strong lysis after entry into stationary phase (see Extended Data Fig. 9d,e). Experimentally, lysis for the SGLC occurred at lower overall AHL concentrations than for the ungated circuit, consistent with the results of the gated RFP test circuit. At AHL concentrations that were effective, the SGLC indeed caused rapid onset of lysis after normal exponential growth, whereas initial exponential growth was slowed by the ungated circuit (Fig. 3e). To quantify these differences, we determined the exponential growth rate at low OD, which was substantially less affected in the gated lysis circuit compared with the ungated circuit (Fig. 3f). At the same time, the SGLC achieved high lysis efficiency—defined as the reduction in OD 1 h after the peak—at much lower AHL concentrations, creating an inducer regime in which strong lysis coexists with minimal

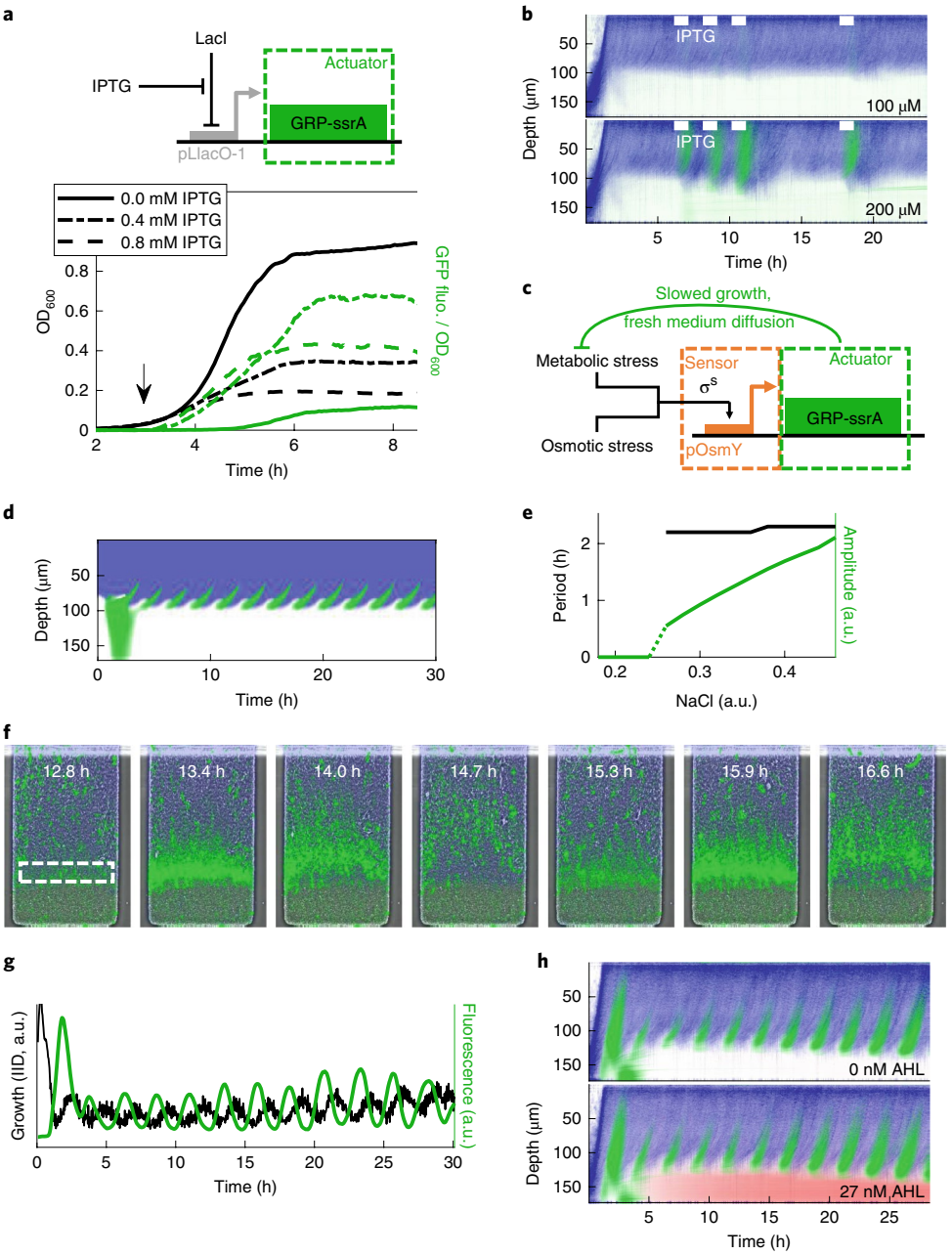


Fig. 2 | Actuator growth modulation and spatiotemporal feedback. a, Schematic of the actuator circuit expressing GRP-ssrA from the pLlacO-1 promoter. The circuit was induced with IPTG at an OD_{600} of 0.03 (black arrow; see Extended Data Fig. 4a for induction at OD_{600} 0.13). **b**, Kymographs showing growth (IIDs, blue) and GFP fluorescence (green) of *E. coli* cells with the actuator circuit in microfluidics. Pulses of IPTG shown by white marks. (See Extended Data Fig. 4 for time-lapse microscopic images and behaviour across several tested IPTG concentrations and for a continuous induction of the actuator circuit.) Note that even though GRP-ssrA is expressed from the same IPTG-inducible promoter as RFP (see Fig. 1g), its relative protein levels across the different phenotypes are not comparable to those of RFP, because the balance of protein synthesis, degradation and dilution is altered by the growth-modulating effect of GRP and the presence of the ssrA-degradation tag. **c**, Sensor-actuator circuit. The pOsmY stress sensor drives the GRP-ssrA actuator protein, giving rise to spatiotemporal negative feedback mediated by growth suppression that increases available nutrient concentration and relieves metabolic stress. **d**, Simulated kymograph from the spatiotemporal model of microcolony dynamics with the sensor-actuator circuit. After sensor and actuator components were separately tuned to match batch and microfluidic experiments with isolated circuits (**a**,**b** and Fig. 1e), the full model predicts periodic oscillations of actuator protein concentration (green) and growth (blue) near the no-growth boundary (see Extended Data Fig. 5 and Supplementary Discussion 2). **e**, Period and amplitude of actuator protein concentration oscillations in the spatiotemporal simulations. **f**, Time-lapse images of cells carrying the sensor-actuator circuit showing oscillations of GRP-ssrA fluorescence (green) at the growth boundary in medium supplemented with 71 mM NaCl needed for feedback activation (compare Extended Data Fig. 6 and see Supplementary Video 4). **g**, Local oscillations in GRP-ssrA fluorescence and growth (calculated via IIDs, see Methods) averaged over the area marked in **f**. **h**, Kymographs showing periodic oscillations of GRP-ssrA fluorescence (green) and growth (IIDs, blue) in a microfluidic colony of cells with the sensor-actuator circuit and expressing untagged RFP (red) from an AHL-inducible pLuxI promoter.

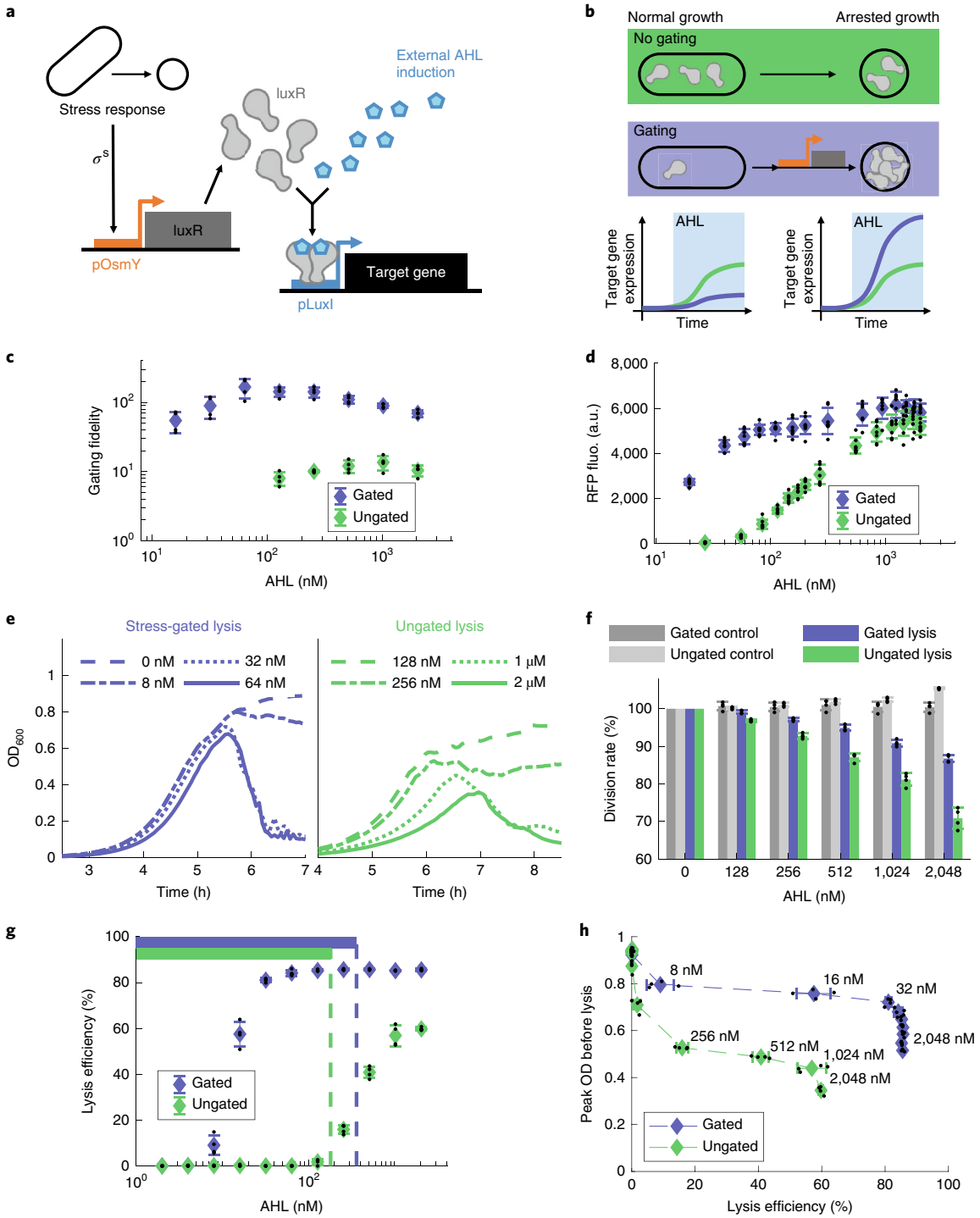


Fig. 3 | Synthetic stress gate primes cells for phenotype-specific expression. a, Stress-gated expression of a target gene. The pOsmY stress sensor drives expression of the regulator protein LuxR required to activate the pLuxI promoter on binding of external AHL. **b**, Constitutive LuxR expression (green) from the native pLuxR promoter in the standard pLuxI induction cassette (compare Supplementary Fig. 4) leads to sensitivity of both growing and growth-arrested phenotypes to pLuxI induction by external AHL. The gating module (purple) reduces LuxR levels during normal growth and primes cells with LuxR on transition into the non-growing phenotype, increasing pLuxI promoter sensitivity and thus target gene expression on induction with external AHL in a phenotype-specific manner. **c**, Stress-gated RFP circuit (target gene = RFP-ssrA-AAV) with constant AHL induction in the plate reader compared with an ungated circuit. Gating fidelity is calculated by dividing the peak RFP fluorescence (compare Extended Data Fig. 8b) by the RFP fluorescence at OD 0.7. Data shown are for peak fluorescence values >5% of peak fluorescence at maximum induction. **d**, The same circuits cultured in microfluidics and induced for 10 h at different AHL levels. Average RFP fluorescence in the growth-arrested region was determined as in Fig. 1h. Plot shows mean ± s.d. ($n = 7$ microfluidic traps). **e**, Batch growth of cells with the stress-gated lysis circuit (**a** with target gene = E-ssrA-AAV) and the ungated circuit at various inducer concentrations. **f**, Growth rates for the stress-gated and ungated lysis circuit, and corresponding controls expressing RFP-ssrA-AAV rather than E-ssrA-AAV. Plot shows mean ± s.d. ($n = 4$ wells). **g**, Lysis efficiency (percentage OD reduction within 1 h after reaching peak OD) for the two lysis circuits. Horizontal bars at the top of the plot indicate the inducer range where the growth rate reduction in exponential phase is <5%. **h**, Peak OD reached before lysis versus lysis efficiency for the gated and the ungated lysis circuit. The s.d. of peak ODs (vertical axis) is smaller than plot symbol sizes. Plots in **c**,**g**,**h** show mean ± s.d. ($n = 4$ wells).

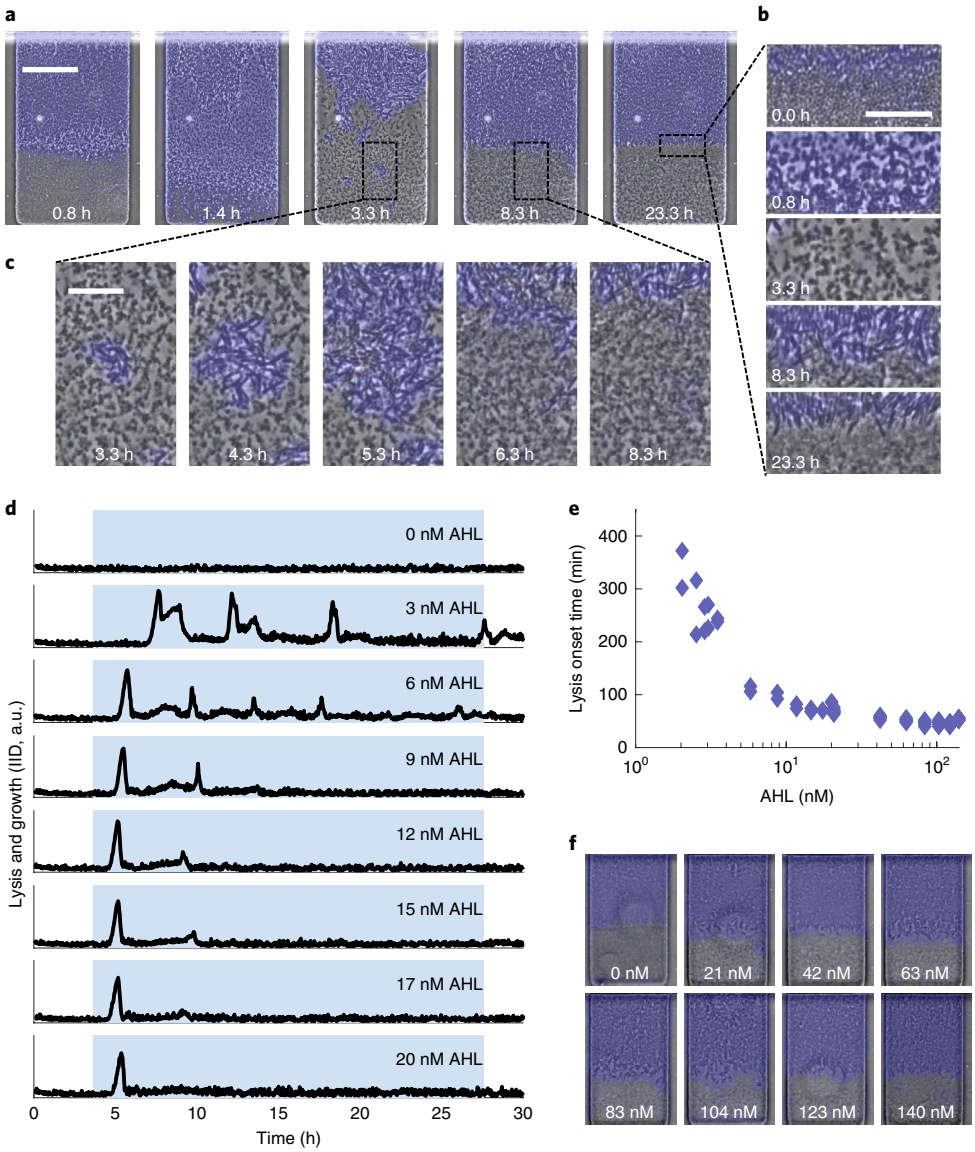


Fig. 4 | A colony-level phenotype filter. a, Time-lapse images of a single microfluidic trap with cells carrying the SGLC (see Supplementary Video 6). Dormant cells are eliminated from the population whereas dividing cells are unaffected by the SGLC. Blue regions based on IIDs (see Methods) indicate regions of active growth and lysis. Time stamps indicate time after beginning of 17 nM AHL induction. Scale bar, 50 µm. **b**, Magnification of the interface region for the same time points as in **a**. Note the similarity of the growth pattern (including the growth boundary) after reaching steady state in **a** and **b** to the undisturbed pattern (see Fig. 1b and Extended Data Fig. 1). Dividing cells are found in the same region of the trap, whereas dormant cells have been eliminated by the SGLC. Scale bar, 15 µm. **c**, Magnification of a region in the rear of the trap, showing a small population of cells growing after the first lysis event, which eventually lyses when steady-state growth in the front part of the trap is established. Scale bar, 15 µm. **d**, Growth and lysis dynamics across AHL inducer levels for the SGLC. The curves were obtained by averaging IIDs (see Methods) in the region of the trap usually occupied by growth-arrested cells. Spikes correspond to sudden lysis events. Blue-shaded regions denote AHL induction windows. **e**, Time from induction to the onset of lysis was determined by detecting the first peak in the IID traces as in **d**. Experimental data were pooled from three different experiments for 0–3.5 nM, 0–20 nM and 0–140 nM AHL. Each data point corresponds to a single trap on the microfluidic chip. **f**, Effective stress gating by the SGLC enables specific elimination of dormant cells even for substantially higher AHL concentrations (compared with the minimum concentration necessary to induce killing; see Extended Data Fig. 10a), resulting in growth patterns very similar to those in **a**. Shown are time-lapse images of the steady state taken after 18 h of AHL induction with the indicated concentration. Growing cells are found in the same region as for the 0 nM control, whereas dormant cells are eliminated for all non-zero AHL concentrations (see Supplementary Video 8 for details).

growth rate reduction in exponential phase (Fig. 3g; compare Fig. 3e). Similar behaviour was observed for the peak OD of the growth curves: in the non-gated circuit, effective stationary-phase lysis was observed only at the cost of substantial peak-OD reduction, whereas, for the SGLC, lysis efficiency was generally high and the change in peak OD was small over more than two orders of magnitude of AHL concentrations (Fig. 3h), indicating effective gating.

In our diffusion-limited microcolonies, the SGLC achieved specific, long-term elimination of growth-arrested cells. In contrast to the ungated circuit, lysis occurred only in the dormant cells. No lysis was observed in the dividing subpopulation (Fig. 4a and see Supplementary Video 6). The steady-state growth structure superficially resembled the undisturbed, natural population structure (compare Fig. 1b); however, the sharp boundary in this case (Fig. 4b)

does not separate growing cells from dormant cells, but from cell debris produced by previous lysis events. During continued exposure to sufficiently high AHL levels (above approximately 9 nM), any cells transitioning to the non-dividing phenotype in the region with inadequate nutrient supply were rapidly eliminated (Fig. 4c). AHL levels mainly determined the time until sufficient lysis protein E had accumulated for the onset of lysis (Fig. 4d, and see Extended Data Fig. 10b and Fig. 4e). At low AHL concentrations, this increased time to lysis led to oscillations, characterized by repeated trap fillings alternating with lysis events several hours after growth arrest (Fig. 4d, and see Extended Data Fig. 10c and Supplementary Video 7).

The eventual steady-state phenotype patterns resulting from inducer levels up to 15-fold higher than the 9 nM minimum were very similar (Fig. 4f, and see Extended Data Fig. 10a and Supplementary Video 8), each consisting only of dividing cells, the growth of which was largely unaffected by the SGLC. This robustness is a direct result of the high gating fidelity and the lysis efficiency of the SGLC observed in batch culture over a wide range of AHL concentrations (compare Fig. 3), and demonstrates the high degree of growth-state specificity of the SGLC.

The expanding toolkit of synthetic biology enables engineering of microbes with increasingly layered and complex functionality[2,33], including population-level spatial patterns based on explicit cell-to-cell communication systems[15–17]. In contrast, our complementary tools are designed to operate in the presence of a pre-existing spatial structure resulting from nutrient deprivation in confined environments. In this study, we demonstrated the construction of synthetic circuits geared specifically towards sensing and modulating this structure by pursuing two control strategies: (1) endogenously controlling cellular metabolism to modify the underlying transport processes, and (2) incorporating growth-state-induced signals into the synthetic regulatory network to achieve phenotype-specific effects or eliminate specific undesired phenotypes.

Due to the ubiquity of confinement and emergent structure in bacterial populations, we envision that the strategies presented in this study will be useful and, in some cases, necessary, to achieve desired circuit behaviour in applications. One important application for circuits with growth state awareness is the design of bacteria-based biosensors, for example, in the context of in vivo disease diagnosis and environmental monitoring[34,35]. Growth-arrested cells in regions with poor nutrient supply could substantially distort the readout signal (a catalytically active enzyme or fluorescent protein) due to its accumulation in the absence of dilution. Using the GRP actuator to induce growth or elimination of those cells with the SGLC would allow the biosensor to reset without replacing the entire cell population. More generally, eliminating host cell phenotypes for which a synthetic circuit was not originally designed could substantially increase its reliability and reduce cell-to-cell variability. Stress-gated lysis circuits could also have specific advantages for targeted drug delivery[36]: triggering lysis by changes in cell growth state would allow cells to produce maximum payload before releasing the drug into the environment, eliminating the need for fine-tuning quorum-sensing-based systems[8]. Growth pattern engineering can provide methods to control the gut colonization behaviour and microbiome interactions for synthetic probiotic bacteria[7], and improve biocontainment strategies that prevent the undesired accumulation of dormant designer microbes[37]. Exploiting the coexistence of different phenotypes with phenotype-gated circuits might also allow for the engineering of division of labour among isogenic cells.

On a more technical level, our microfluidic chip and its multiplexing capabilities are also an ideal test bed to probe changes in bacterial physiology due to heterogeneous nutrient deprivation in different growth media, and to assess the impact of these changes on the function of synthetic circuits. Although our experimental realization is highly idealized, the artificial microenvironments mimic an aspect of natural environments that is difficult to investigate using traditional microbiological techniques. In contrast to liquid cultures and colonies grown on agar plates, coexisting growth phenotypes can be observed for extended periods of time at single-cell resolution. As in nature, these phenotypes emerge due to the self-generated spatial gradients, without the need to adjust the chemical environment to obtain a particular physiological state[24]. Similar to our observations of expression from the native stress response promoter pOsmY and from synthetic constructs, other transcriptional and proteomic changes can be quantitatively characterized using appropriate reporters. In addition, the population structure provides a direct readout for changes in metabolism and the phenotypic switching behaviour itself (which might be of interest, for example, in the context of metabolic engineering), and thus makes them accessible to quantitative investigation.

Methods

Plasmids and strains. All DNA constructs were created using EMILI[38] (for deletions and short modifications) or Gibson Assembly[39] (for assembly of multiple parts). Templates for PCR were either previous constructs from our group[8,40,41] (inducible promoters and fluorescent proteins) or the *E. coli* MG1655Z1 genome (stress promoters). To target proteins for degradation by the protease ClpXP, strong and weak ssrA-degradation tags[42] with amino acid sequences 'AANDENYALAA' and 'AANDENYAAAV', respectively, were fused to the C-terminus of the corresponding protein, encoded immediately before the stop codon. All constructs were verified by Sanger sequencing.

All strains in this study were created by transforming the host strain MG1655Z1 with plasmids of the ColE1 and p15A family. MG1655Z1 expresses high levels of the repressors TetR and LacI, the latter of which was required for induction experiments with the pLlac-O1 promoter in non-dividing cells (see Fig. 1g) as well as the controlled LuxR expression experiments (see Supplementary Fig. 3a). Therefore, it was used for all experiments for consistency. Supplementary Table 1 lists the strains, ordered by their first appearance, along with the contained plasmids, selection antibiotics (Amp, ampicillin; Cm, chloramphenicol and Kan, kanamycin) and references to figures containing experimental data obtained from them. Maps for all constructs can be found in Supplementary Fig. 4; the corresponding nucleotide sequences are attached as Supplementary Data. Plasmid names use the following conventions: 'LAA' and 'AAV' refer to the strong and weak ssrA-degradation tags with amino acid sequences 'AANDENYALAA' and 'AANDENYAAAV', respectively[42]. The term 'pLux' refers to the standard AHL-inducible expression cassette with divergent expression of LuxR and the protein of interest (compare Supplementary Fig. 4). Constructs missing the LuxR portion of the cassette are labelled 'noLuxR' explicitly. The gating plasmid (bold) enables gated expression from such noLuxR constructs (compare Fig. 3a).

Our initial studies of σ^S-dependent promoters included those for genes *poxB*, *yciG* and *osmY* based on their previously shown strong response to the multiple stresses[27]. We eventually settled on *osmY*, because it showed a particularly peaked response on entry into stationary phase and tunability by NaCl (see Fig. 1e). The essential part of our standard pOsmY promoter was PCR amplified from the genome as nucleotides −235 to 427 relative to the transcriptional start site of *osmY*, based on a previous characterization that showed notably higher expression levels and inducibility for this region compared with truncated versions[43]. A TAA stop codon was appended to terminate the N-terminal region of the OsmY protein contained in this region and a strong ribosome-binding site[28] was inserted immediately after the stop codon to form the stress sensor promoter that we call pOsmY (see Supplementary Fig. 4). The actuator protein GRP was discovered when the N-terminal region of the OsmY protein was instead fused directly to the N-terminus of GFP using a SerAlaGly linker. Although the mechanism of action is not essential for our findings, the growth-modulating effect of GRP is most probably due to the signalling peptide (residues 1–28) that usually targets OsmY to the periplasm[44].

Culture media and inducers. All plate reader and microscopy experiments were performed in EZ-rich medium (Teknova) prepared according to the manufacturer's instructions with 0.2% glucose as the carbon source, except for the gating fidelity tests shown in Extended Data Fig. 8d,e, which were conducted in lysogeny broth (Difco). To prevent cell adhesion to microfluidic channels, all media were supplemented with 0.075% (w:v) Tween 20 (for consistency also in plate reader experiments). Selection antibiotics (50 μg ml^{-1} of ampicillin, 50 μg ml^{-1} of kanamycin and/or 34 μg ml^{-1} of chloramphenicol) and basal inducers were added as required. A stock suspension of 1-μm-diameter fluorescent beads (Polysciences, Fluoresbrite Multifluorescent, catalogue no. 24062, visible in the RFP channel) was created at a ratio of one drop per 250 μl of MilliQ water. The resulting suspension was added to all microfluidic media at a dilution of 1:1,000 to be able to measure flow velocity in the media channels (see Microfluidics and microscopy). The specific AHL used in this study was *N*-(3-oxohexanoyl)-L-homoserine lactone, also known as 3-oxo-C6-HSL or Auto Inducer 1 for *Vibrio fischeri*. Large volumes of chemicals/inducers (for example, 5 M NaCl in MilliQ water used to reach a target

concentration of 500 mM) were added by replacing the corresponding amount of water in the EZ-rich recipe.

Microfluidics and microscopy. Microfluidic experiments were conducted using the techniques established in previous studies from our lab[41,45]. Briefly, polydimethylsiloxane was poured on a silicon wafer mould (created using photolithography) and baked at 80 °C for 1 h before cutting individual chips, punching fluidic access ports and cleaning bonding surfaces. Polydimethylsiloxane pieces containing single chips were then bonded to standard glass cover-slips after a 3-min exposure in a Jelight UV-Ozone cleaner and kept overnight at 80 °C for strong bonding. A schematic of the specific microfluidic device used in this study can be found in Supplementary Fig. 1.

The night before the experiment, a culture of the corresponding strain was inoculated from a −80 °C glycerol stock into lysogeny broth (Difco) supplemented with the appropriate selection antibiotics (see Supplementary Table 1 for a list of strains) and grown overnight in a shaking incubator at 37 °C. On the day of the experiment, the saturated overnight culture was diluted 1:1,000 into 5 ml of the same medium supplemented with 0.075% (w:v) Tween 20 and incubated until an OD_{600} of 0.1 was reached. Then, 1 ml of this culture was spun down at 3,000*g* in a standard tabletop microcentrifuge for 3 min, and the cells were resuspended in 5 µl of the base medium used for the microfluidic experiment (see Culture media and inducers). The cell suspension was then immediately loaded into the waste port of the microfluidic device, which had been degassed in a vacuum chamber for at least 30 min, and base medium was loaded into all other ports. After the traps had been seeded, media and waste ports were connected to the corresponding reservoirs and media channels were flushed with base medium at a high flow rate (>2,000 µm s^{-1}) before calibrating the flow rate to 100 µm s^{-1} in the media channels by measuring the streak length of fluorescent beads at a 1-s exposure in the RFP-imaging channel. A Nikon Eclipse Ti2 inverted microscope was used with a Photometrics CoolSNAP HQ2 CCD camera to capture images and a Lumencor SOLA SE LED Light Engine for fluorescent excitation. The stage holding the microfluidic device was enclosed in a plexiglass incubation chamber, maintaining a constant 37 °C environment. The acquisition software was NIS-Elements.

We performed time-lapse imaging with two settings: at a magnification of ×10, we used exposure times of 500 µs for phase contrast, 500 ms for GFP (25% light-emitting diode (LED) intensity) and 500 ms for RFP (20% LED intensity). All 14 traps for a single medium composition (seven for each of the two sizes) could be captured in a single image, requiring eight images for the whole chip, taken every 4 min. High-resolution data (for example, for analysis of growth patterns as in Fig. 1b,c) was acquired at ×30, with exposure times of 20 ms for phase contrast, 100 ms for GFP (25% LED intensity) and 500 ms for RFP (20% LED intensity). With one image for each trap size and medium composition (16 images), each containing two traps, an imaging interval of 2 min was possible.

Microscopy data analysis. Image registration using customized MATLAB scripts was performed on phase-contrast images to correct for stage jitter. Masks for individual traps and media channels were automatically generated for data extraction. Dye intensity measured in media channels was used to calculate media compositions generated by the microfluidic mixer (see Supplementary Figs. 1 and 2). We created kymographs from microscopy data by averaging intensities from the appropriate imaging channels (or interimage differences (IIDs) for growth, see below) along the horizontal direction of the trap for each time point and concatenating the resulting columns chronologically.

We analysed the growth pattern in the traps by calculating IIDs between consecutive phase-contrast images. Note that IIDs themselves are sensitive to any type of intensity change, be it due to growth, lysis or cell motion. The IID for a pixel at position (*i, j*) in frame *t* of the data was calculated as

$$\mathrm{IID}(i,j,t) = \frac{1}{25}\left|\sum_{k,l\in N_5(i,j)} I(k,l,t) - I(k,l,t-1)\right|, \quad (1)$$

where $N_5(i, j)$ is the 5 × 5 pixel^2 neighbourhood of the pixel (*i, j*), and $I(i, j,t)$ is the intensity value at position (*i, j*) in frame *t*. The local averaging before taking the absolute value was done to account for residual jitter after registration. The raw IIDs were then used to produce a time series by averaging them over the areas indicated in the corresponding figure captions (for example, see Figs. 2g and 4d). Kymographs were calculated from IIDs as described above. To add some dynamic information to static microscopy images and thereby aid their interpretation, we further smoothed the raw IIDs from equation (1) with a 20 × 20 uniform kernel, and overlaid the corresponding phase-contrast images with blue colour where the resulting smoothed IIDs exceeded a specific threshold value. The threshold value for blue colour in microscopy images and kymographs was chosen as the minimum value that reliably excluded non-growing regions of the colony.

Plate reader experiments. Microplate experiments were conducted in standard, flat-bottom, 96-well plates (Falcon) in a Tecan Infinite M200 Pro plate reader at 37 °C. GFP fluorescence was recorded with an excitation wavelength of 485 nm and an emission wavelength of 520 nm, and RFP fluorescence was recorded at 580 nm/620 nm. Both fluorescence and 600 nm absorbance (OD_{600}) values were recorded every 3.5 min. Experiments were started from a 1:1,000 dilution of saturated cultures. Lysis strains were pre-incubated for 1 h before adding inducer to prevent growth delays due to priming with LuxR at the previous growth arrest. A background signal obtained from the first 10 measurements in each well was subtracted from all OD and fluorescence traces. Fluorescence was then divided by OD_{600} to yield a normalized fluorescence signal.

The gating fidelity for the stress-gated RFP circuit (see Fig. 3c and Extended Data Fig. 8d,e) was calculated as the ratio between the peak normalized RFP fluorescence signal for the well and the normalized fluorescence at OD 0.7 (see Extended Data Fig. 8a). Lysis efficiency was defined as the percentage reduction in OD 1 h after reaching the peak OD. Exponential cell doubling rates (see Fig. 3f and Supplementary Fig. 3) were calculated as the slope of the least-squares linear fit to $\log_2(\mathrm{OD}(t))$ between an OD_{600} of 0.08 and one of 0.2.

Numerical simulations. Chemical reaction kinetics of the sensor, actuator and gating components, along with population dynamics and nutrient consumption, were simulated using custom scripts written in MATLAB. For the spatiotemporal model, we constructed a one-dimensional reaction–diffusion–advection model using a split-step, finite-difference, implicit-diffusion numerical scheme. A detailed description of the model equations, assumptions and parameters can be found in Supplementary Discussion 2.

Statistics and reproducibility. Behaviour similar to the examples shown in Fig. 1b,c and Extended Data Fig. 1a,b was observed at the start of virtually every microfluidic experiment with standard EZ growth medium ($n \geq 10$) in all microfluidic traps, before any structure-modulating gene circuits were chemically induced.

The examples in Fig. 1e represent two replicate wells on the same plate and two independent plate reader experiments.

The examples in Fig. 1f and Extended Data Fig. 3a,b show typical behaviour observed, for each concentration, in two microfluidic traps in the same experiment and across two experiments.

The data in Fig. 1g,h and Extended Data Fig. 3c represent averages from seven separate microfluidic traps for each concentration, extracted from a single microfluidic experiment for each plot.

The examples shown in Fig. 2a and Extended Data Fig. 4a represent two replicate wells on the same plate and are consistent with the behaviour observed in three independent plate reader experiments with varying IPTG induction levels and time points.

The examples shown in Fig. 2b and Extended Data Fig. 4b,c represent the behaviour recorded in two microfluidic traps in the same experiment for each concentration.

The examples shown in Fig. 2f–h and Extended Data Fig. 6a represent the behaviour recorded in two microfluidic traps in the same experiment for each concentration and across three or more independent experiments.

The examples in Fig. 3e represent the behaviour recorded in four replicate wells on the same plate (corresponding to the data in Fig. 3f–h), as well as of two independent plate reader experiments.

The examples shown in Fig. 4a–f and Extended Data Fig. 10 represent the behaviour recorded for two or more microfluidic traps for each concentration, and consistent with the behaviour observed in three separate microfluidic experiments for varying AHL concentrations. The examples shown in Supplementary Fig. 2b,c represent all microfluidic experiments that made use of the on-chip mixing feature of the microfluidic design ($n \geq 10$).

The data shown in Extended Data Fig. 2a,b represent two microfluidic traps for each concentration in the same experiment.

The data shown in Extended Data Fig. 4d,e represent the behaviour recorded in two microfluidic traps for each concentration in the same experiment and consistent across five more IPTG concentrations not shown.

The data shown in Extended Data Fig. 6c represent two wells in the same plate reader experiment.

The examples shown in Extended Data Fig. 7a,b,c represent the behaviour recorded in two microfluidic traps for each concentration in the same experiment and three independent experiments with varying AHL concentrations.

The examples shown in Extended Data Fig. 8a represent the behaviour recorded in four replicate wells on the same plate and two independent plate reader experiments.

The experiments shown in Supplementary Fig. 3a,b,c were conducted once. The growth rate dependence on LuxR content in Supplementary Fig. 3a was observed consistently across 5 (for no LuxR/pLlacO-1-LuxR) or 12 (for constant LuxR/pOsmY-LuxR) wells of different AHL concentrations. The data for the pLlacO-1-RFP in Supplementary Fig. 3c represent two replicate wells on the same plate.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are either attached as source data or available from the corresponding authors upon request. The blueprints of the

microfluidic device necessary to recreate the specific microenvironments used in this study are also available from the corresponding authors upon request.

Code availability

The modelling code for the numerical simulations and the scripts used to analyse experimental data are available from the corresponding authors upon request.

Received: 5 February 2019; Accepted: 7 February 2020;
Published online: 13 April 2020

References

1. Riglar, D. T. & Silver, P. A. Engineering bacteria for diagnostic and therapeutic applications. *Nat. Rev. Microbiol.* **16**, 214–225 (2018).
2. Bittihn, P., Din, M. O., Tsimring, L. S. & Hasty, J. Rational engineering of synthetic microbial systems: from single cells to consortia. *Curr. Opin. Microbiol.* **45**, 92–99 (2018).
3. Ye, H. et al. Self-adjusting synthetic gene circuit for correcting insulin resistance. *Nat. Biomed. Eng.* **1**, 0005 (2017).
4. Nissim, L. et al. Synthetic RNA-based immunomodulatory gene circuits for cancer immunotherapy. *Cell* **171**, 1138–1150 (2017).
5. Gupta, A., Reizman, I. M. B., Reisch, C. R. & Prather, K. L. J. Dynamic regulation of metabolic flux in engineered bacteria using a pathway-independent quorum-sensing circuit. *Nat. Biotechnol.* **35**, 273–279 (2017).
6. Cremer, J., Arnoldini, M. & Hwa, T. Effect of water flow and chemical environment on microbiota growth and composition in the human colon. *Proc. Natl Acad. Sci. USA* **114**, 6438–6443 (2017).
7. Dou, J. & Bennett, M. R. Synthetic biology and the gut microbiome. *Biotechnol. J.* **13**, e1700159 (2017).
8. Din, M. O. et al. Synchronized cycles of bacterial lysis for in vivo delivery. *Nature* **536**, 81–85 (2016).
9. Lyssiotis, C. A. & Kimmelman, A. C. Metabolic interactions in the tumor microenvironment. *Trends Cell Biol.* **27**, 863–875 (2017).
10. Agapakis, C. M., Boyle, P. M. & Silver, P. A. Natural strategies for the spatial optimization of metabolism in synthetic biology. *Nat. Chem. Biol.* **8**, 527–535 (2012).
11. Belkin, S. et al. Remote detection of buried landmines using a bacterial sensor. *Nat. Biotechnol.* **35**, 308–310 (2017).
12. Phelan, V. V., Liu, W.-T., Pogliano, K. & Dorrestein, P. C. Microbial metabolic exchange-the chemotype-to-phenotype link. *Nat. Chem. Biol.* **8**, 26–35 (2012).
13. Liu, J. et al. Metabolic co-dependence gives rise to collective oscillations within biofilms. *Nature* **523**, 550–554 (2015).
14. Cao, Y. et al. Collective space-sensing coordinates pattern scaling in engineered bacteria. *Cell* **165**, 620–630 (2016).
15. Grant, P. K. et al. Orthogonal intercellular signaling for programmed spatial behavior. *Mol. Syst. Biol.* **12**, 849 (2016).
16. Kong, W., Blanchard, A. E., Liao, C. & Lu, T. Engineering robust and tunable spatial structures with synthetic gene circuits. *Nucleic Acids Res.* **45**, 1005–10014 (2017).
17. Toda, S., Blauch, L. R., Tang, S. K. Y., Morsut, L. & Lim, W. A. Programming self-organizing multicellular structures with synthetic cell-cell signaling. *Science* **361**, 156–162 (2018).
18. Stewart, P. S. & Franklin, M. J. Physiological heterogeneity in biofilms. *Nat. Rev. Microbiol.* **6**, 199–210 (2008).
19. Flemming, H.-C. et al. Biofilms: an emergent form of bacterial life. *Nat. Rev. Microbiol.* **14**, 563–575 (2016).
20. Besharova, O., Suchanek, V. M., Hartmann, R., Drescher, K. & Sourjik, V. Diversification of gene expression during formation of static submerged biofilms by *Escherichia coli*. *Front. Microbiol.* **7**, 1568 (2016).
21. Storz, G. & Hengge, R. *Bacterial Stress Responses* (American Society for Microbiology Press, 2010).
22. Davies, J. A. Real-world synthetic biology: is it founded on an engineering approach, and should it be? *Life* **9**, 6 (2019).
23. Scott, M., Gunderson, C. W., Mateescu, E. M., Zhang, Z. & Hwa, T. Interdependence of cell growth and gene expression: origins and consequences. *Science* **330**, 1099–1102 (2010).
24. Gefen, O., Fridman, O., Ronin, I. & Balaban, N. Q. Direct observation of single stationary-phase bacteria reveals a surprisingly long period of constant protein production activity. *Proc. Natl Acad. Sci. USA* **111**, 556–561 (2014).
25. Lange, R. & Hengge-Aronis, R. Growth phase-regulated expression of bolA and morphology of stationary-phase *Escherichia coli* cells are controlled by the novel sigma factor sigma-S. *J. Bacteriol* **173**, 4474–4481 (1991).
26. Mather, W., Mondragón-Palomino, O., Danino, T., Hasty, J. & Tsimring, L. S. Streaming instability in growing cell populations. *Phys. Rev. Lett.* **104**, 208101 (2010).
27. Weber, H., Polen, T., Heuveling, J., Wendisch, V. F. & Hengge, R. Genome-wide analysis of the general stress response network in *Escherichia coli*: σ^S-dependent genes, promoters, and sigma factor selectivity. *J. Bacteriol.* **187**, 1591–1603 (2005).
28. Lutz, R. & Bujard, H. Independent and tight regulation of transcriptional units in *Escherichia coli* via the LacR/O, the TetR/O and AraC/I1-I2 regulatory elements. *Nucleic Acids Res.* **25**, 1203–1210 (1997).
29. Kim, Y.-I., Burton, R. E., Burton, B. M., Sauer, R. T. & Baker, T. A. Dynamics of substrate denaturation and translocation by the ClpXP degradation machine. *Mol. Cell* **5**, 639–648 (2000).
30. Marguet, P., Tanouchi, Y., Spitz, E., Smith, C. & You, L. Oscillations by minimal bacterial suicide circuits reveal hidden facets of host-circuit physiology. *PLoS ONE* **5**, e11909 (2010).
31. Scott, S. R. et al. A stabilized microbial ecosystem of self-limiting bacteria using synthetic quorum-regulated lysis. *Nat. Microbiol.* **2**, 17083 (2017).
32. Shadel, G. S. & Baldwin, T. O. Positive autoregulation of the *Vibrio-fischeri* luxR gene. *J. Biol. Chem.* **267**, 7696–7702 (1992).
33. Ozdemir, T., Fedorec, A. J. H., Danino, T. & Barnes, C. P. Synthetic biology and engineered live biotherapeutics: toward increasing system complexity. *Cell Syst.* **7**, 5–16 (2018).
34. Sedlmayer, F., Aubel, D. & Fussenegger, M. Synthetic gene circuits for the detection, elimination and prevention of disease. *Nat. Biomed. Eng.* **2**, 399–415 (2018).
35. Roggo, C. & van der Meer, J. R. Miniaturized and integrated whole cell living bacterial sensors in field applicable autonomous devices. *Curr. Opin. Biotechnol.* **45**, 24–33 (2017).
36. Claesen, J. & Fischbach, M. A. Synthetic microbes as drug delivery systems. *ACS Synth. Biol.* **4**, 358–364 (2015).
37. Chan, C. T. Y., Lee, J. W., Cameron, D. E., Bashor, C. J. & Collins, J. J. 'Deadman' and 'passcode' microbial kill switches for bacterial containment. *Nat. Chem. Biol.* **12**, 82–86 (2016).
38. Füzik, T., Ulbrich, P. & Ruml, T. Efficient mutagenesis independent of ligation (EMILI). *J. Microbiol. Methods* **106**, 67–71 (2014).
39. Gibson, D. G. et al. Enzymatic assembly of DNA molecules up to several hundred kilobases. *Nat. Methods* **6**, 343–345 (2009).
40. Danino, T., Mondragón-Palomino, O.,Tsimring, L. & Hasty, J. A synchronized quorum of genetic clocks. *Nature* **463**, 326–330 (2010).
41. Prindle, A. et al. A sensing array of radically coupled genetic 'biopixels'. *Nature* **481**, 39–44 (2012).
42. Andersen, J. B. et al. New unstable variants of green fluorescent protein for studies of transient gene expression in bacteria. *Appl. Environ. Microbiol.* **64**, 2240–2246 (1998).
43. Yim, H. H., Brems, R. L. & Villarejo, M. Molecular characterization of the promoter of osmY, an rpoS-dependent gene. *J. Bacteriol.* **176**, 100–107 (1993).
44. Yim, H. H. & Villarejo, M. osmY, a new hyperosmotically inducible gene, encodes a periplasmic protein in *Escherichia coli*. *J. Bacteriol.* **174**, 3637–3644 (1992).
45. Stricker, J. et al. A fast, robust and tunable synthetic gene oscillator. *Nature* **456**, 516–519 (2008).

Acknowledgements

The microfluidic mixer is derived from a design developed by M. Dueck. We also thank G. Graham, R. Cooper and R. Golestanian for stimulating discussions and Q. Tran for technical help. This material is based on work supported by the National Institutes of Health/National Institute of General Medical Sciences (grant no. RO1-GM069811) and the National Science Foundation (grant no. DMS-1463657). P.B. was supported by the Human Frontiers Science Program fellowship LT000840/2014-C. Any opinions, findings and conclusions or recommendations expressed in this material are those of the authors(s) and do not necessarily reflect the views of the funding agencies.

Author contributions

P.B. and A.D. created the stress promoter construct and discovered the growth modulation feedback in microfluidic experiments. P.B., A.D., L.S.T. and J.H. designed the study. P.B. designed the microfluidic chip, generated the remaining constructs and strains, performed the plate reader and microscopy experiments, and analysed the data. P.B. and L.S.T. carried out the numerical simulations. P.B., A.D., L.S.T. and J.H. interpreted the results and wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41564-020-0686-0.

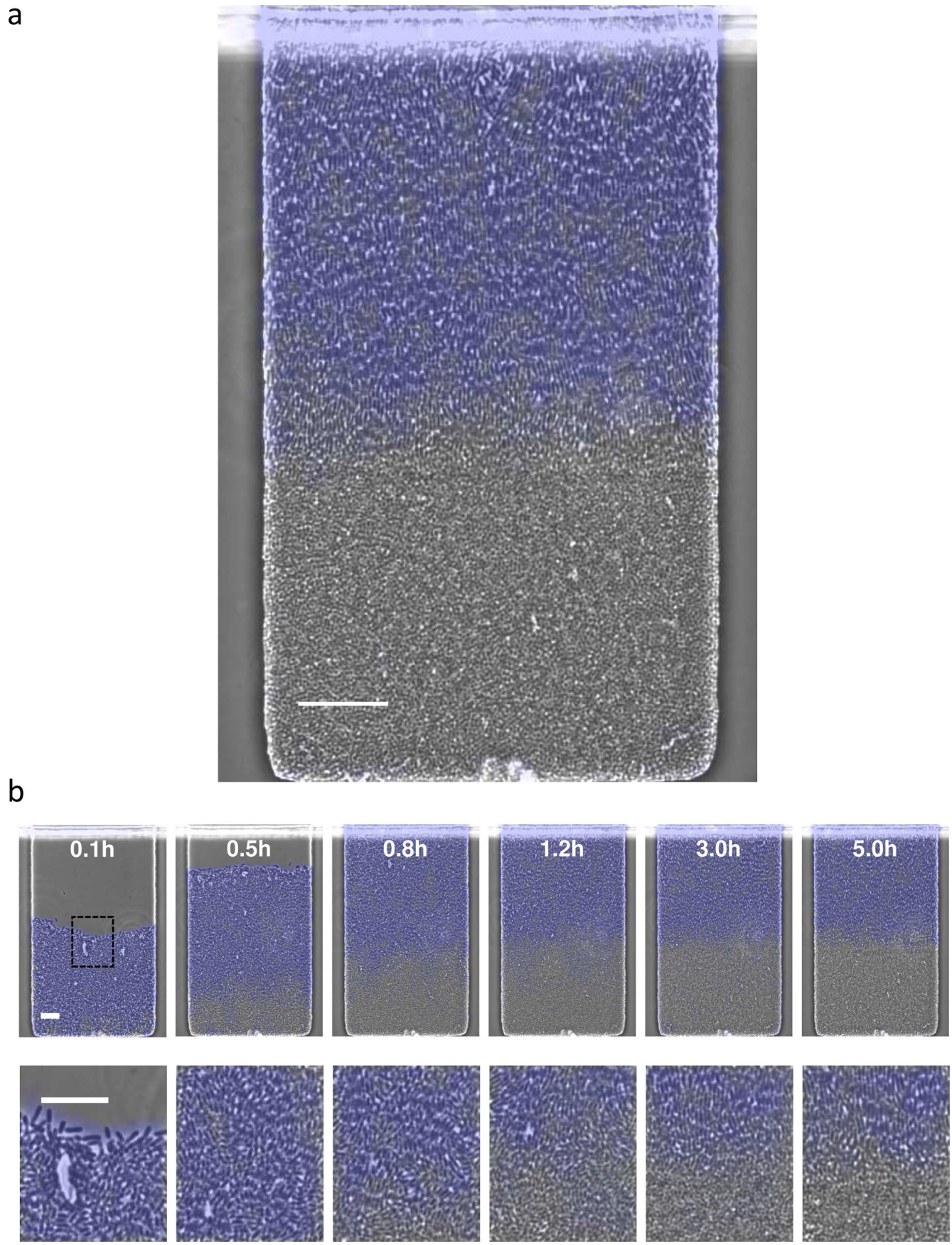
Supplementary information is available for this paper at https://doi.org/10.1038/s41564-020-0686-0.

Correspondence and requests for materials should be addressed to L.S.T. or J.H.

Reprints and permissions information is available at www.nature.com/reprints.

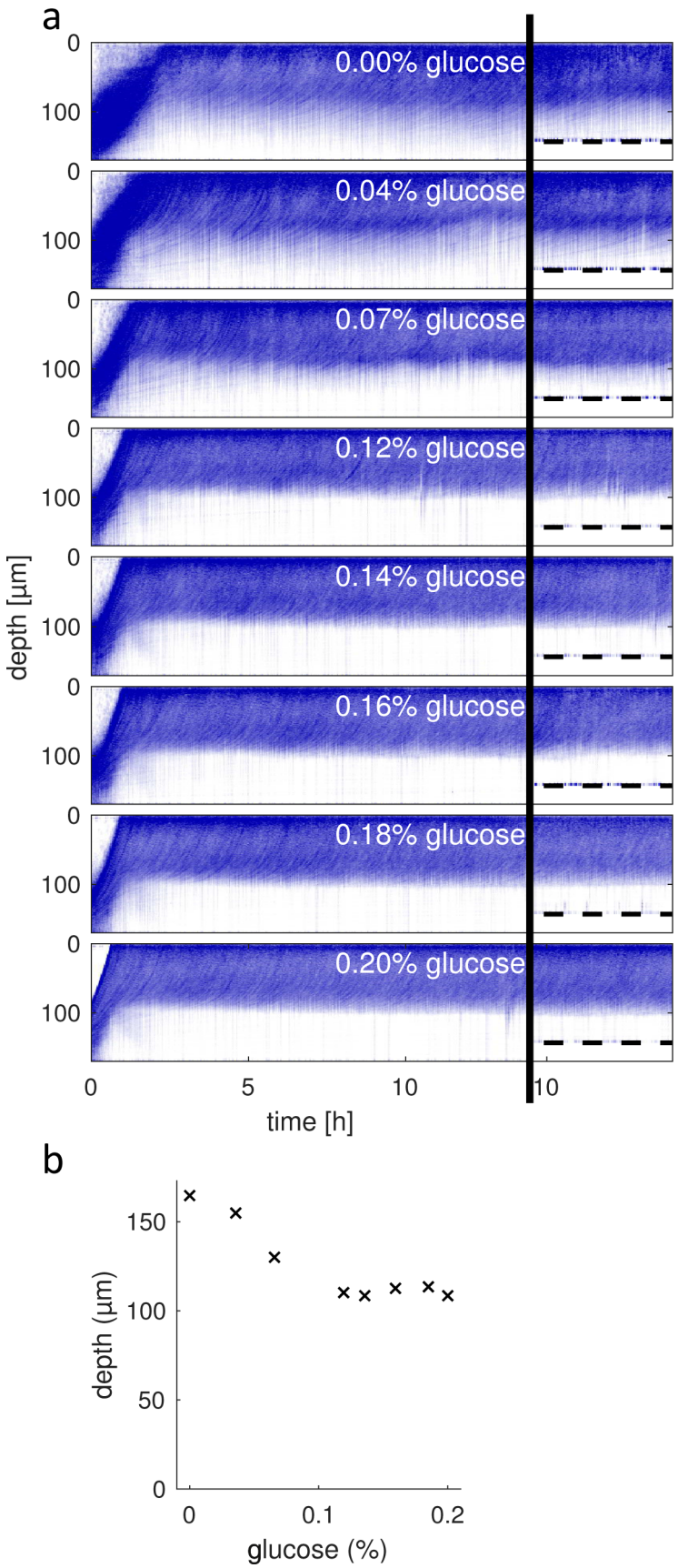
Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

© The Author(s), under exclusive licence to Springer Nature Limited 2020



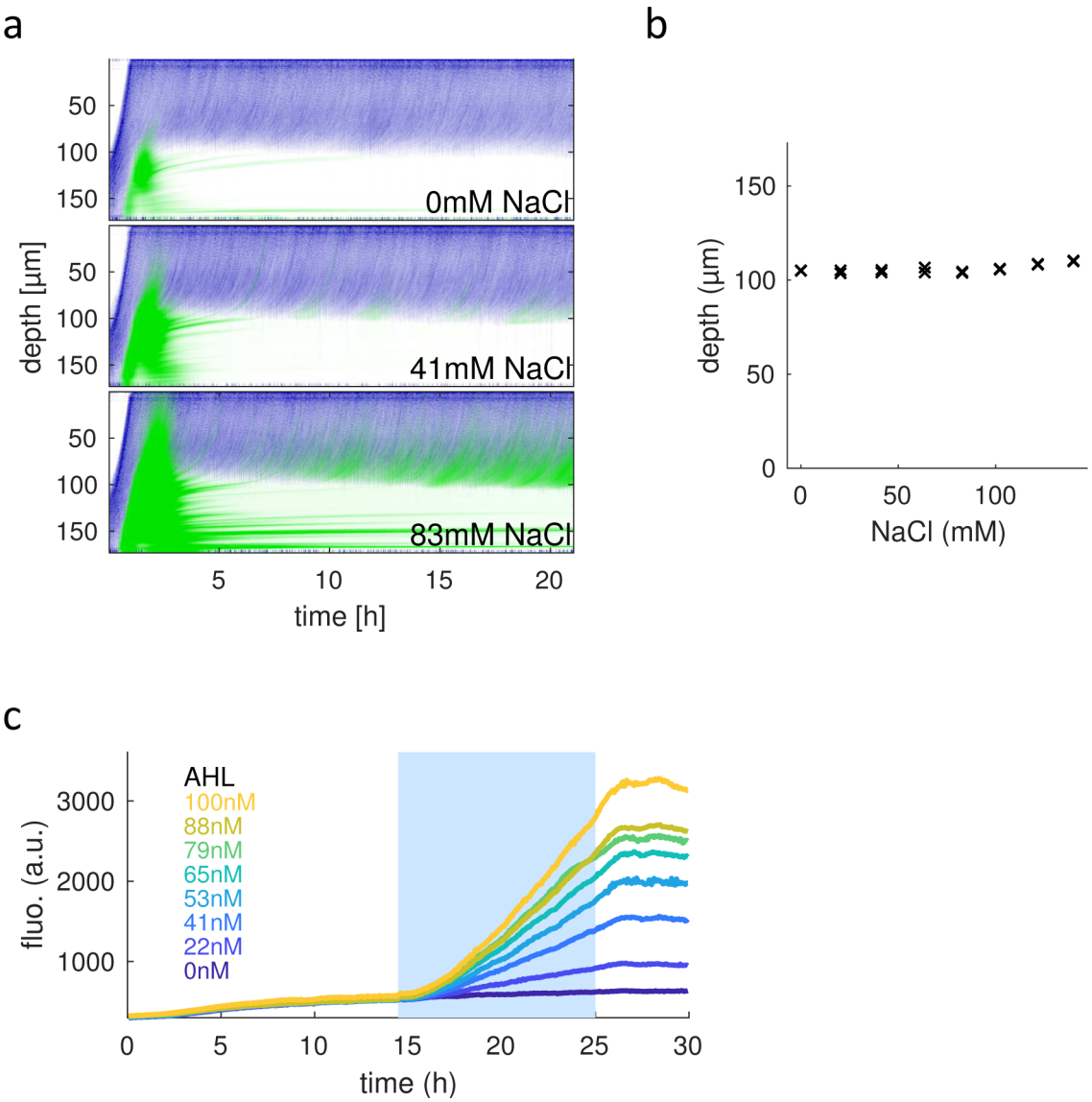
Extended Data Fig. 1 | See next page for caption.

Extended Data Fig. 1 | Establishment of phenotypic heterogeneity. a, High-resolution image of full microfluidic trap after establishment of phenotypic heterogeneity. Blue color indicates IIDs (see Methods). Scale bar, 20μm. **b**, Initial growth in the microfluidic trap. Time-lapse images as in Fig. 1b with zoomed-in region around the eventual growth boundary (see also Supplementary Video 1). Blue color indicates IIDs (see Methods). Scale bars in top left and bottom left image, 15μm.

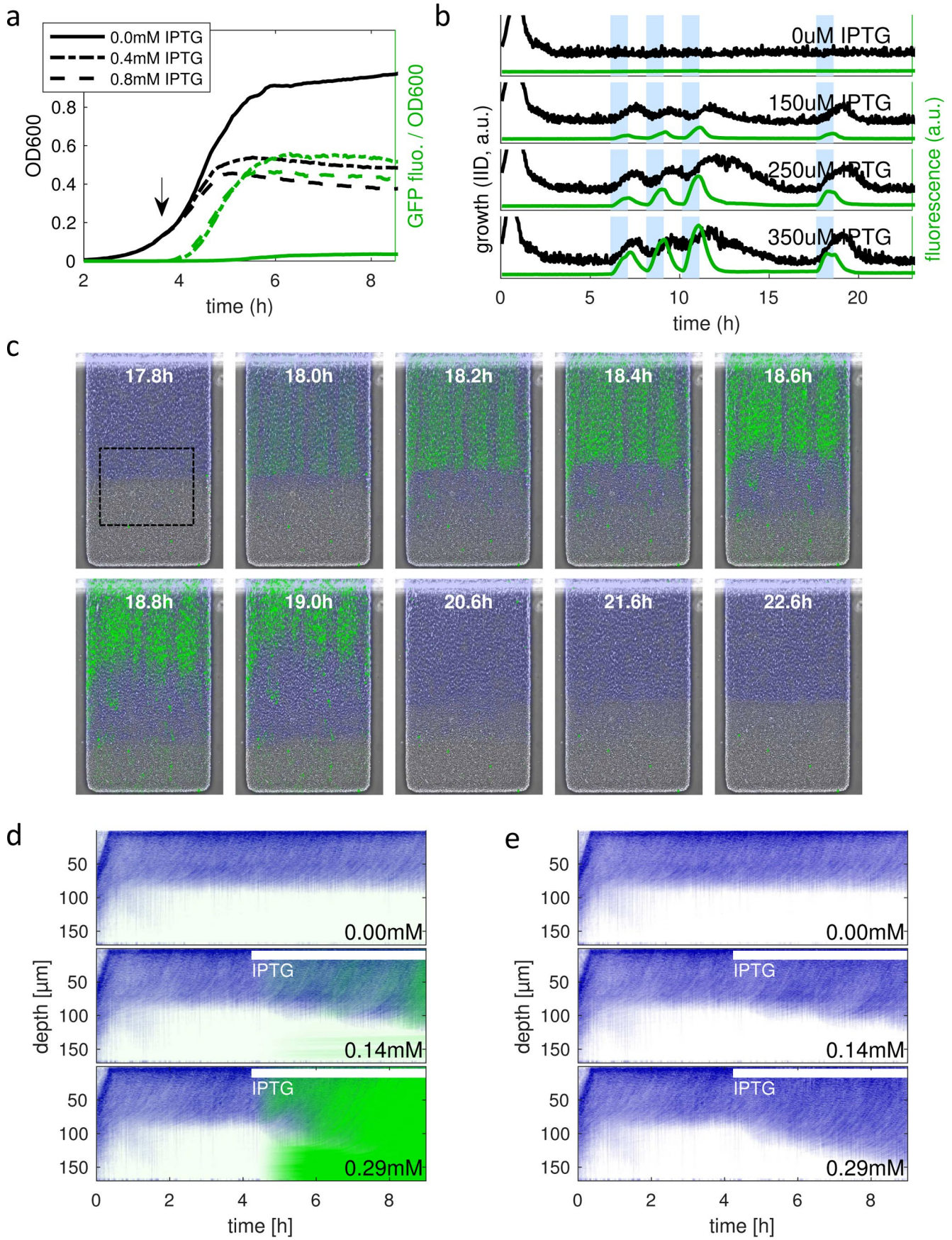


Extended Data Fig. 2 | See next page for caption.

Extended Data Fig. 2 | Growth patterns in different media compositions. a, Kymographs indicating inter-image differences (IIDs, see Methods) as in Fig. 1c during the establishment of a steady state growth pattern in 170μm deep traps for growth media containing different amounts of glucose (the standard concentration is 0.2% w/v). For lower concentrations, growth extends further into the back of the trap with a smoother transition between regions of growth and no growth. Kymographs on the right of the black line show consistent behavior in the smaller (140μm deep) traps (the dashed line indicates the end of the trap). **b**, Distance from the mouth of the microfluidic trap at which the inter-image difference (IID) shown in **a** drops below detectable levels. For this purpose, the IID profile was averaged over 4 h of steady-state growth. The threshold for growth detection was chosen as the lowest possible level that safely detected the no-growth region in standard growth medium.

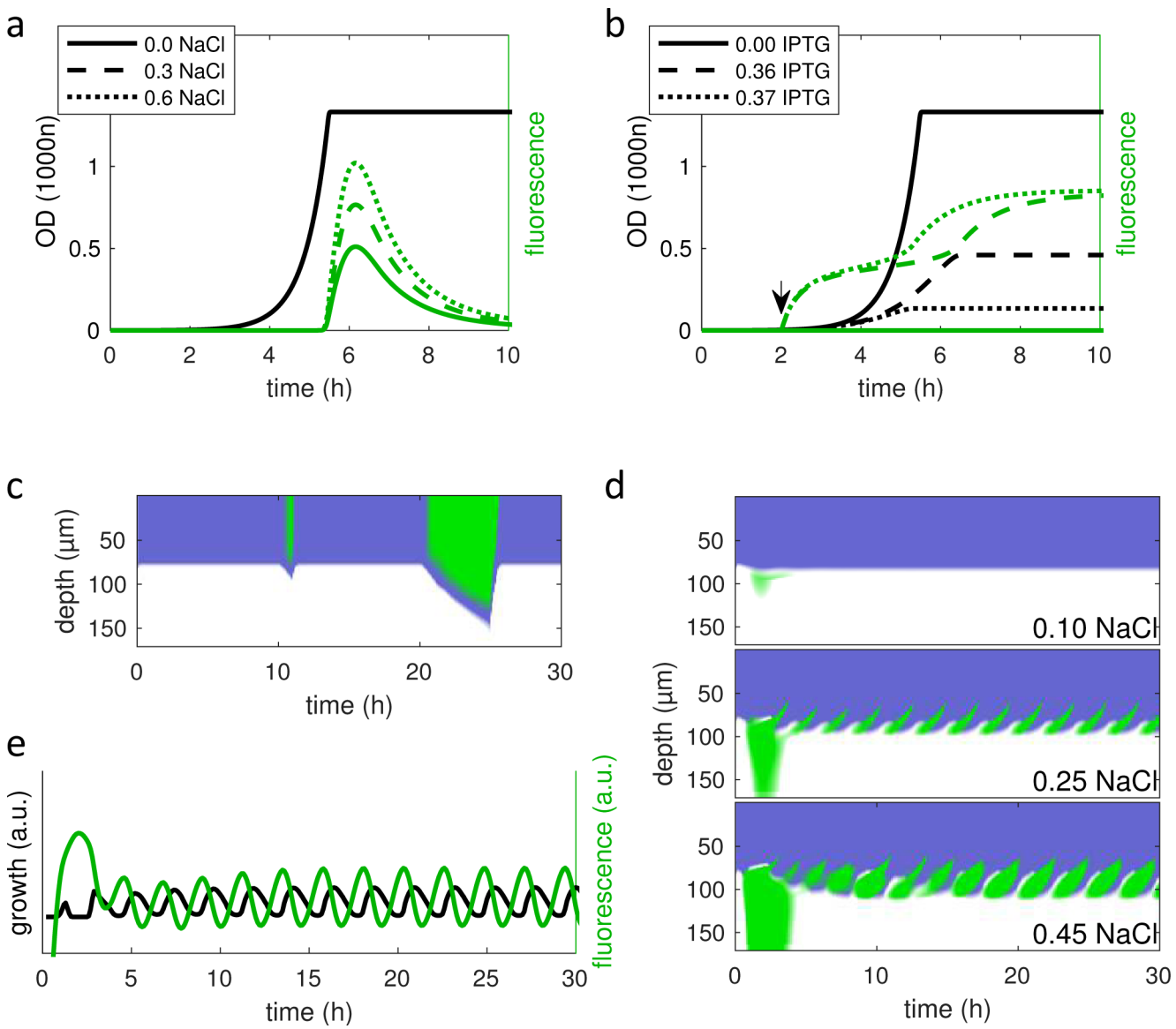


Extended Data Fig. 3 | Transition to dormancy and protein expression in dormant cells. a, Stress sensor activation at different background osmolarities. Kymographs showing growth (IIDs, blue) and GFP fluorescence (green) for lower NaCl concentrations. Compared to Fig. 1f, the fluorescence signal has been rescaled to visualize even weak activation of the pOsmY stress sensor. While the sensor is transiently activated upon growth arrest even in the absence of basal osmotic stress (0 mM NaCl), higher NaCl also elicits a measurable response in dividing cells close to the growth boundary. **b**, Position of the growth interface during sensor activation, analyzed as in Extended Data Fig. 2b, showing no detectable growth modulation by NaCl itself or the activation of the sensor module. See Extended Data Fig. 6c for additional controls. **c**, Expression of untagged RFP from the pLuxI promoter in growth-arrested cells. As in Fig. 1g, h, the plot shows average fluorescence measured in the growth-arrested back of the microfluidic trap. The blue shaded area marks the window of induction with AHL.

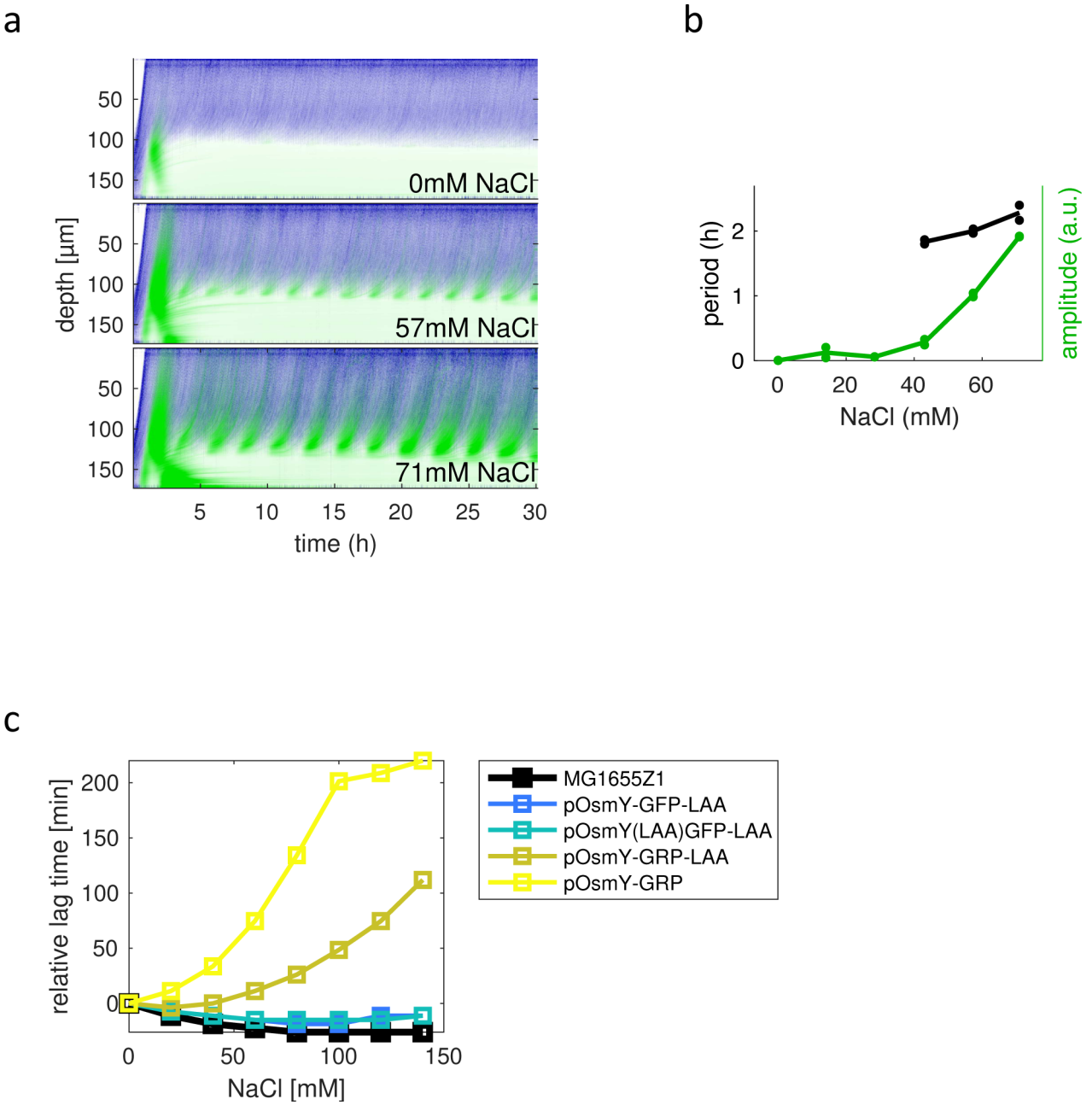


Extended Data Fig. 4 | See next page for caption.

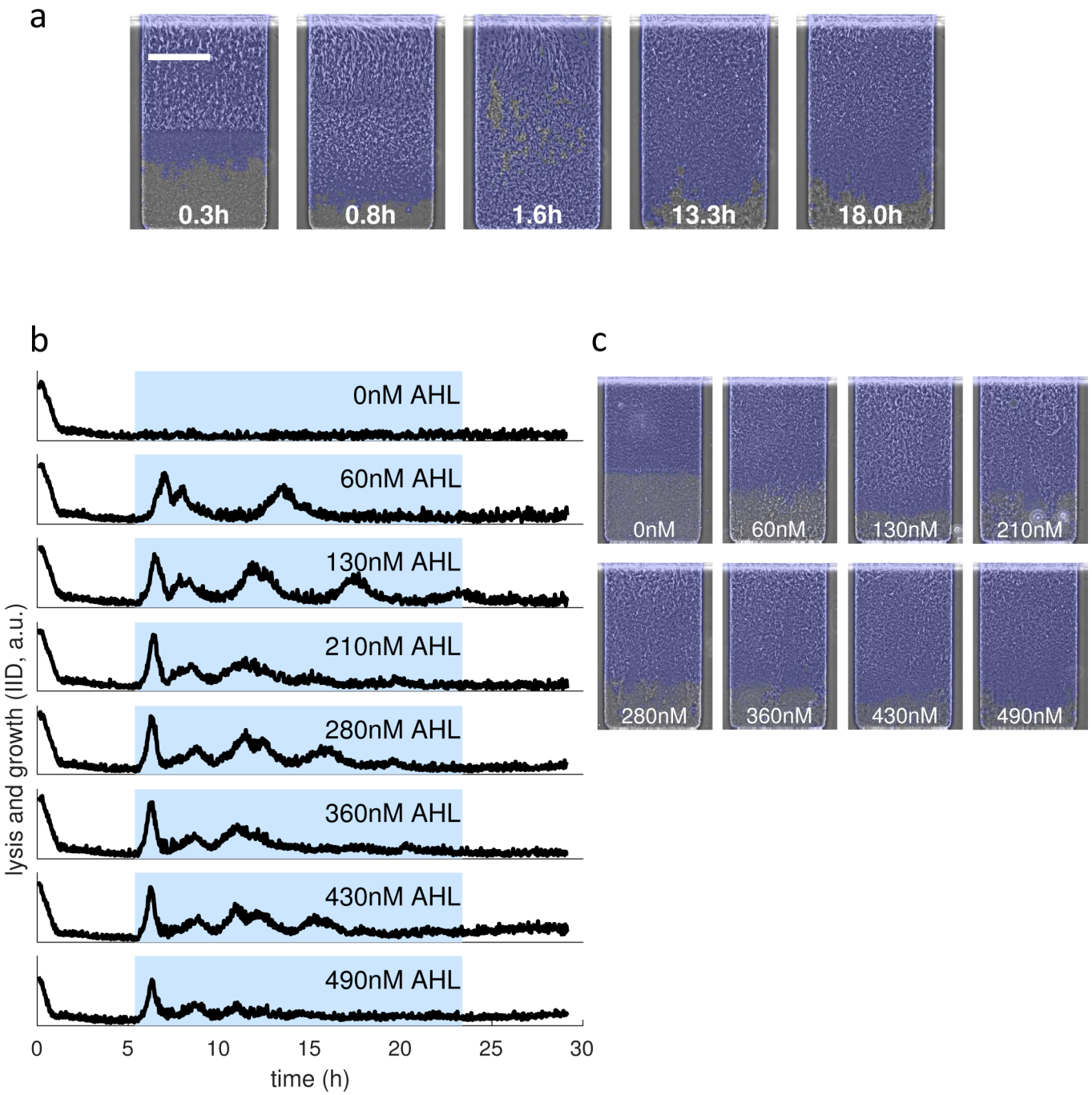
Extended Data Fig. 4 | Actuator circuit in batch culture and microfluidics. a, IPTG was added when the culture of *E. coli* cells equipped with the actuator circuit reached an OD600 of 0.13. Growth is halted through expression of GRP-ssrA from pLlacO-1 at a higher OD compared to Fig. 2a, consistent with the later addition of IPTG. **b**, Time series of growth (calculated as inter-image difference, IID, see Methods) and fluorescence in our microfluidic environment, averaged over the area in the trap indicated in **c**. Blue shaded regions mark the induction windows for all 8 IPTG concentrations tested. Cumulative effects can be observed when pulses of IPTG are given in short succession. **c**, Time-lapse images showing growth (IIDs, blue) and fluorescence (green) for the experiment shown in panel b during the last pulse of 300 μM IPTG (see Supplementary Video 3). The pulse begins at $t = 17.6$ h. **d**, Kymographs of growth (IIDs, blue) and fluorescence (green) for cell populations carrying the actuator circuit, which is induced continuously starting around $t = 4$ h. With increasing GRP-ssrA expression, the area of active growth extends further and further into the trap. **e**, Same data as in **d**, omitting fluorescence for better visualization of the growth pattern.



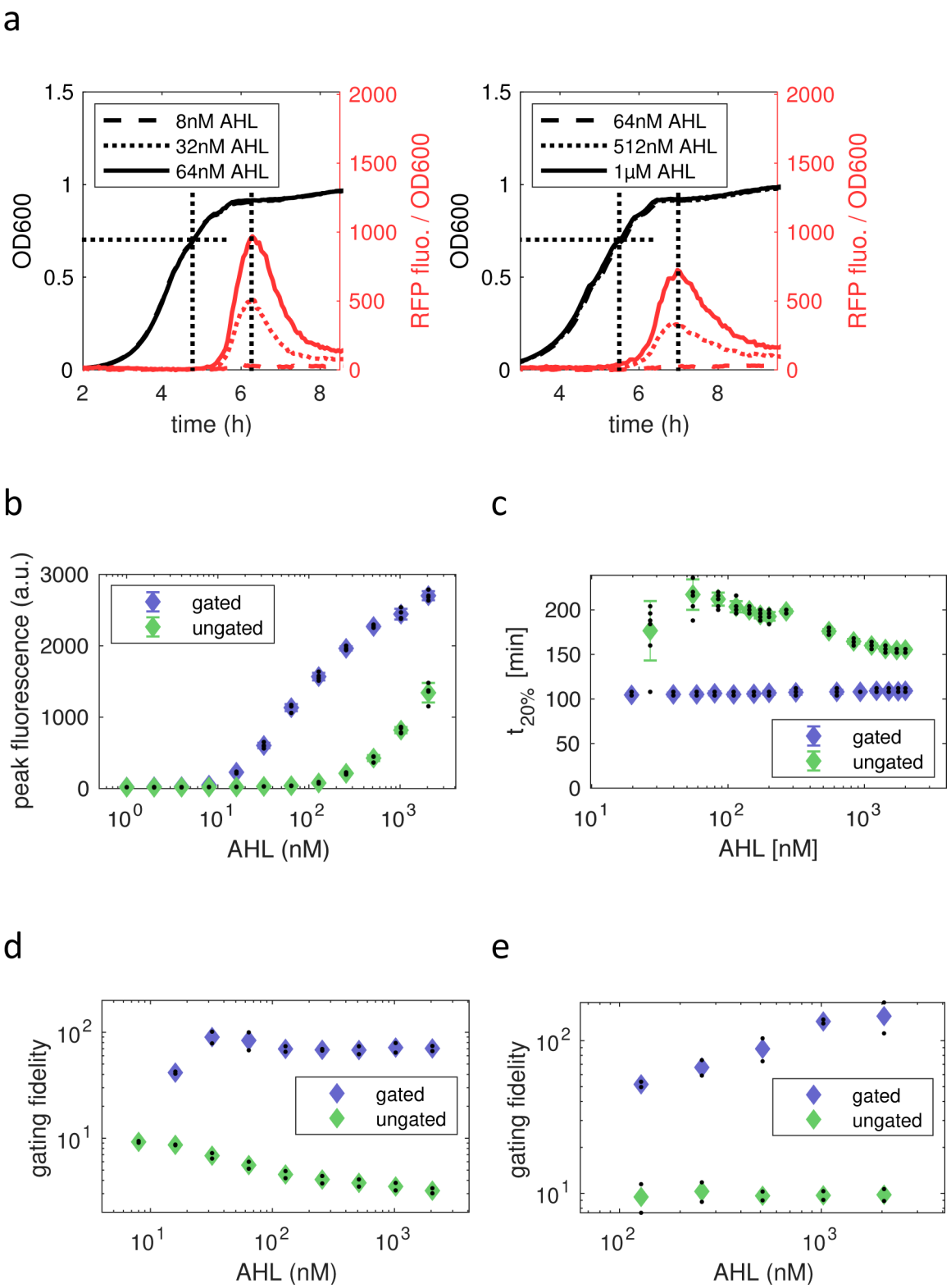
Extended Data Fig. 5 | Numerical simulations of the sensor-actuator circuit (see Supplementary Discussion 2 for modeling details). a, Activation of the stress sensor upon growth arrest in numerical simulations of a well-mixed batch culture of cells. The promoter strength of the sensor is modeled to increase with the NaCl concentration in the media (cf. Figure 1e). **b**, Actuator and population dynamics in batch culture upon IPTG induction. The arrow marks the start of induction. The production rate of the actuator protein is proportional to the IPTG induction level, halting growth earlier for higher IPTG (cf. Figure 2a). **c**, Spatiotemporal simulations of actuator protein induction and population dynamics in a microfluidic trap. 1 h and 5 h pulses of IPTG (amplitudes 0.5) lead to reversible growth resumption in the growth arrested region of the simulated microcolony (cf. Figure 2b and Extended Data Fig. 4). **d**, Numerical simulations of the full spatiotemporal model of the sensor-actuator circuit coupled with population and nutrient dynamics. Oscillations near the growth interface are observed for sufficiently high induction (NaCl) levels of the sensor promoter. **e**, In the oscillatory layer of the microcolony (compare **d**), induction of the actuator protein is followed by growth resumption, which is followed by growth arrest and induction of the actuator protein from the stress sensor, restarting the cycle (cf. Figure 2g).



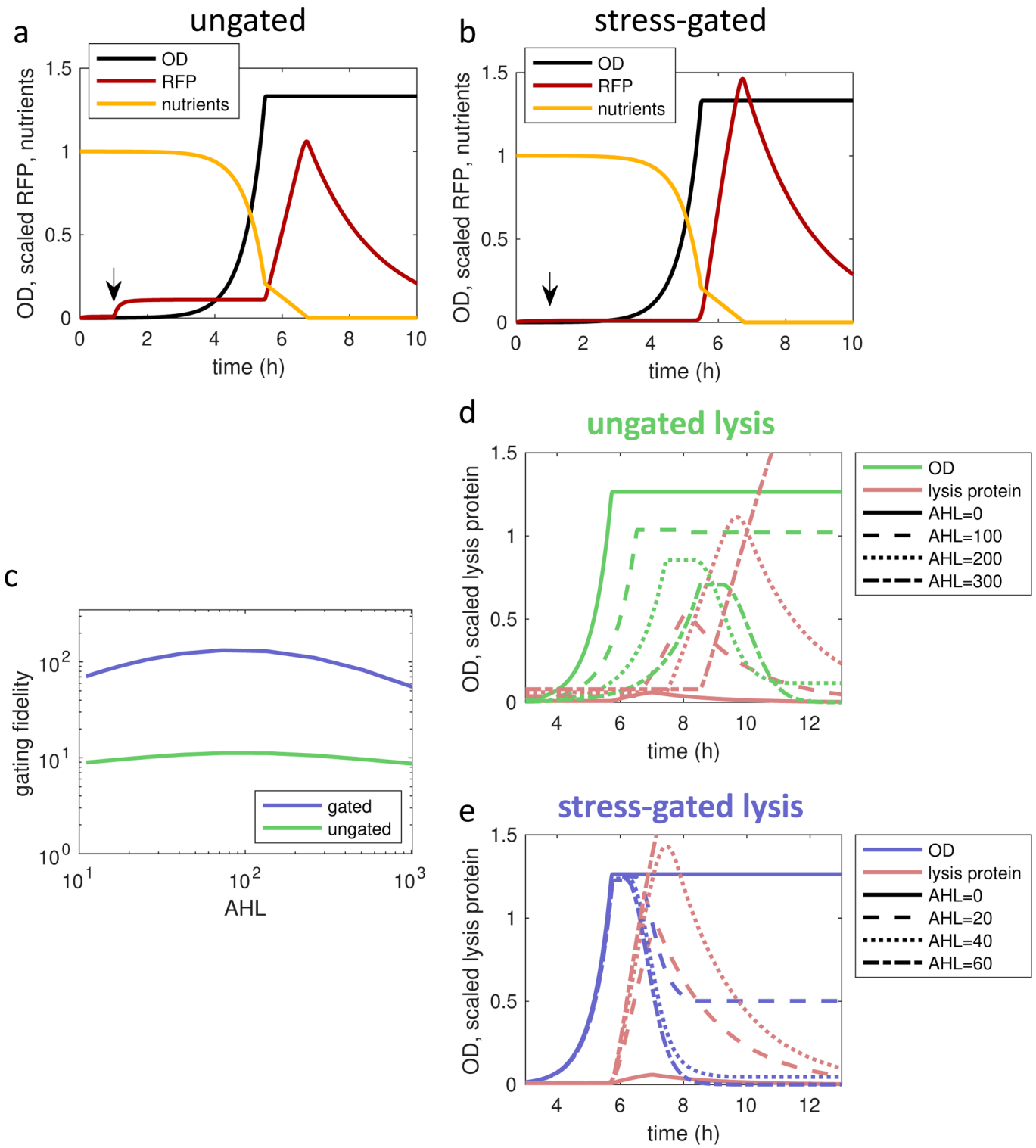
Extended Data Fig. 6 | Induction of oscillations in the sensor-actuator circuit. a, Kymographs of growth (IIDs, blue) and fluorescence (green) for populations carrying the sensor-actuator circuit (Fig. 2c). Basal osmotic stress (NaCl) activates the sensor close to the interface (Fig. 1f) and initiates negative feedback leading to oscillations. **b**, Period and amplitude of fluorescence oscillations. Period is only shown for NaCl concentrations yielding finite oscillation amplitudes. Data points for each concentration correspond to two traps imaged at high resolution (see Methods). **c**, NaCl-dependent growth delay (measured as the time to reach OD 0.1 relative to 0 mM NaCl) in the sensor-actuator circuit (pOsmY-GRP-LAA) and multiple control strains: MG1655Z1 (no construct), pOsmY-GFP-LAA (the sensor component), pOsmY(LAA)-GFP-LAA (the sensor component with an additional ssrA-LAA degradation tag on the N-terminal OsmY part of pOsmY) and pOsmY-GRP (the sensor-actuator circuit without the ssrA-LAA degradation tag on GRP). The presence of growth delays exclusively in the GRP-containing constructs (exacerbated when the ssrA degradation tag is not present and GRP levels are increased) shows that the N-terminal portion of the OsmY protein (no matter whether expressed at high or low levels) has no growth-modulating effect for the media, NaCl concentrations and resulting expression levels relevant for this study (cf. Extended Data Fig. 3b). Note that these additional control constructs are not included in Supplementary Figure 4.



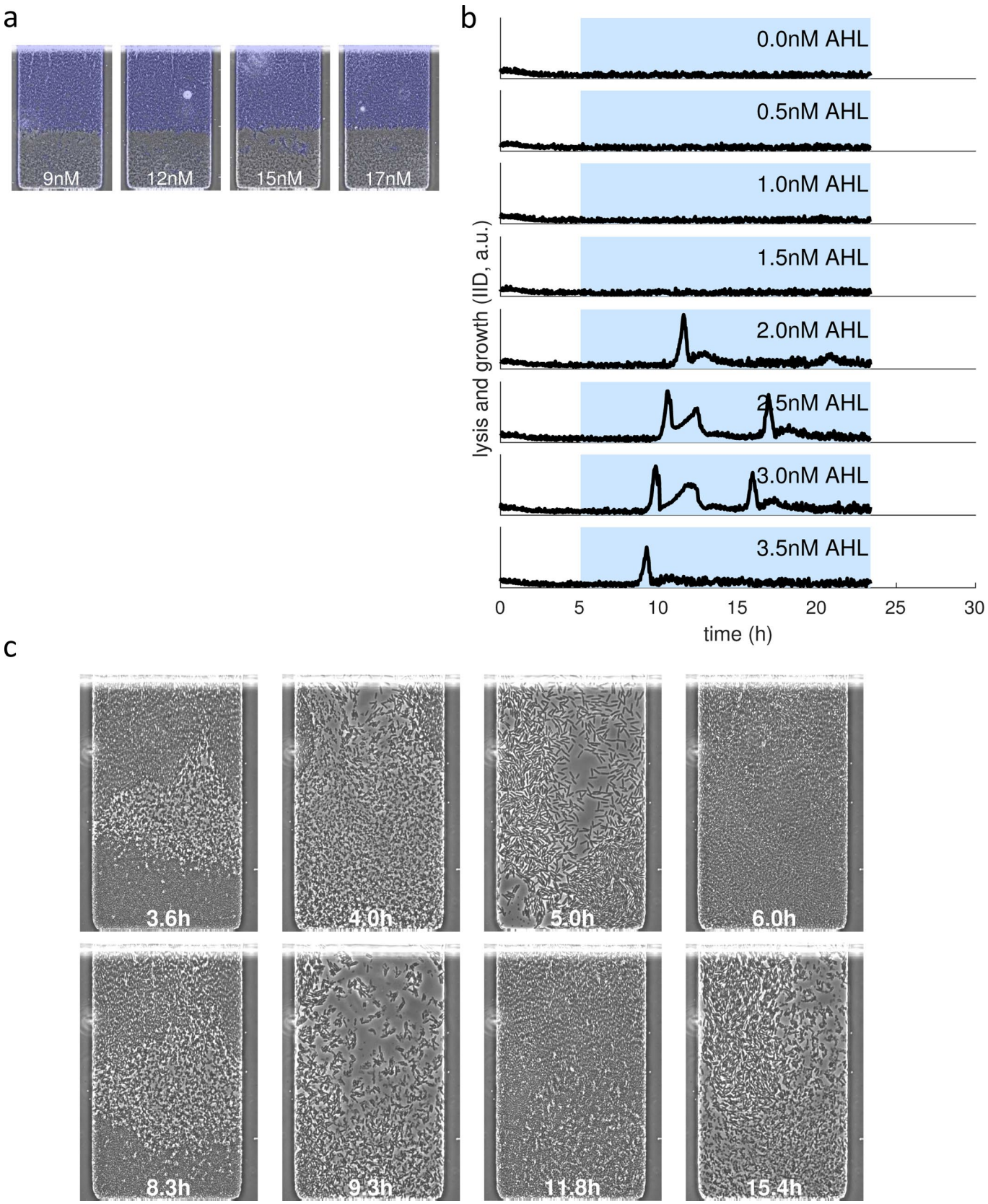
Extended Data Fig. 7 | Unsuccessful elimination of growth-arrested cells with the ungated lysis circuit. a, Time-lapse images of unsuccessful phenotype elimination with the ungated lysis circuit (LuxR-pLuxR-pLuxI-E-ssrA) for induction with 490 nM AHL (compare **b**,**c**). Lysis starts in the growing part of the population, causing growth-arrested cells to receive fresh nutrients and resume growth, while simultaneously also lysing. Cells regrowing despite circuit activation form disorganized colonies and display no visible phenotype pruning. Time stamps denote time after beginning of induction. Blue color indicates inter-image differences (IIDs, see Methods). See also Supplementary Video 5. Scale bar, 50μm. **b**, Growth and lysis dynamics (calculated via inter-image differences, IIDs, see Methods) in the region of the trap usually occupied by growth-arrested cells for different AHL induction levels of the ungated lysis circuit. After the initial lysis, no sharp lysis events are observed. Instead, uncoordinated phases of spatially inhomogeneous lysis, regrowth and growth arrest without lysis lead to fluctuating IID that settles at a low level (cf. end states in **c**). **c**, Images taken after 18 h of AHL induction with the indicated concentration, when traps have settled into a steady state. Blue color indicates inter-image differences (IIDs, see Methods). While cells in all traps initially lyse (**b**), no distinct phenotype elimination is observed and colony structure is disorganized (see Supplementary Discussion 1).



Extended Data Fig. 8 | Stress-gated expression of RFP. a, Example traces of OD and RFP fluorescence in plate reader experiments with the gating circuit (Fig. 3a) with RFP-ssrA-AAV as the target gene (*left*), compared to ungated expression from a standard bidirectional LuxR-pLuxR-pLuxI cassette with constitutive LuxR (*right*). Plotted induction levels were chosen to show similar levels of expression. **b**, Peak fluorescence in plate reader experiments of the same RFP-ssrA-AAV gating circuit (Fig. 3a). The stress-gated circuit exhibits increased sensitivity at low concentrations, which enabled us to determine gating fidelity (Fig. 3c) for much lower concentrations for this circuit compared to constitutive LuxR. Plot shows mean ± *s.d.*, $n = 4$ wells. **c**, We tested the same circuits in microfluidics and measured the fluorescence in the growth-arrested region in the back of the trap in response to a 10 h induction with different AHL concentrations. $t_{20\%}$ is the time until 20% of peak fluorescence (as shown in Fig. 3d) are reached. Plot shows mean ± *s.d.*, $n = 7$ microfluidic traps. **d**, Gating fidelity for stress-gated and ungated RFP-ssrA-AAV from plate reader experiments as in Fig. 3c, but measured in LB. **e**, Gating fidelity for stress-gated and ungated RFP-ssrA-AAV from plate reader experiments as in Fig. 3c, but measured in LB supplemented with 0.2% glucose. Symbols in **d, e** represent means of 2 wells each.



Extended Data Fig. 9 | Numerical simulations of the gating circuit (see Supplementary Discussion 2 for modeling details). a, Time traces for OD, nutrients and RFP as the target protein in the ungated circuit with constitutive production of LuxR. The arrow marks the beginning of AHL induction. **b**, Same time traces for the gated circuit with LuxR expressed from pOsmY, showing reduced expression during exponential phase. **c**, Gating fidelity (defined the same way as for Fig. 3c and Extended Data Fig. 8d,e) for the gated and ungated circuits, showing a ca. 10-fold stronger preference for stationary-phase expression in the gating circuit compared the ungated circuit. **d**, Time traces of OD and cellular lysis protein concentration in the ungated circuit for different AHL levels. Exponential growth rate is impacted at the same time as stationary-phase lysis becomes effective. **e**, Same time traces for the gated circuit for different AHL levels. The reduction of LuxR during exponential phase and its increase in stationary phase leads to efficient lysis without impacting exponential growth.



Extended Data Fig. 10 | See next page for caption.

Extended Data Fig. 10 | Behavior of the SGLC at low AHL concentrations near onset. a, Images taken after 24 h of AHL induction with the indicated concentration, when traps have settled into a steady state. Blue color indicates inter-image differences (IIDs, see Methods). Lower concentrations than those shown here do not settle into steady states (see **c**) or show imperfect pruning. **b**, Growth and lysis dynamics (calculated via inter-image differences, IIDs, see Methods) in the region of the trap usually occupied by growth-arrested cells. No lysis occurred below AHL concentration of 2 nM. For higher concentrations and lysis onset times across parameters, see Fig. 4d–f. **c**, Time-lapse images showing oscillatory behavior of the SGLC close to onset of lysis (3 nM AHL, cf. panel **b** and Fig. 4d). Long delay between LuxR priming by the gating circuit and accumulation of sufficient lysis protein (Fig. 4e) causes sequential fill-up and lysis (see Supplementary Video 7). Time stamps denote time after beginning of induction.

nature research

Corresponding author(s): Jeff Hasty

Last updated by author(s): Jan 10, 2020

Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see Authors & Referees and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	The image acquisition software was "NIS Elements", version 4.10.01. The plate reader data acquisition software was "Tecan i-control", versions 1.11.1.0 and 1.12.4.0.
Data analysis	The analysis of optical and plate reader data was performed using standard data processing steps which are fully described in the Methods section. The mathematical model and its parameters are fully described in the supplement. Our implementation in MATLAB is available from the corresponding authors upon request.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

The data that support the findings of this study (Figs. 1-4, Extended Data Figs. 1-10, Supp. Figs. 1-4) are either provided with the manuscript as source data or available from the corresponding authors upon request.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical methods were used to predetermine sample size. Sample size was determined by experimental bandwidth and was sufficient according to the reproducibility of the data.
Data exclusions	No data were excluded from the analyses.
Replication	Sequences of newly assembled plasmids were verified by Sanger sequencing. Microfluidic and microplate reader experiments relevant for the main findings of the paper were repeated. All attempts at replication were successful.
Randomization	As described in the Methods section, bacteria for experiments where sampled from overnight or diluted liquid cultures of strains with known identity. Thus, selection of individual bacteria was inherently random. Beyond this, there was no group allocation and therefore randomization was not relevant to this study.
Blinding	Investigators were not blinded to strain identity. However, as all data acquisition was automated and identical data processing pipelines were used for different strains and conditions, bias due to non-blinding was not present.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging